

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.6	8	3.2	0.3	0.3	73.6
U2 - Beta(0.5, 0.5)	16.1	8	2.1	0.5	0.3	73
U2 - Beta(0.7, 0.3)	16.6	8.4	1.8	0.1	0.2	72.9
U2 - Beta(1, 1)	16.1	8	1.8	0.3	0.1	73.7
U3 - Beta(0.3, 0.7)	17.7	7.8	1.4	0.2	0.1	72.8
U3 - Beta(0.5, 0.5)	18.5	7.2	1.1	0.2	0.2	72.8
U3 - Beta(0.7, 0.3)	18.1	6.6	1.4	0.2	0.2	73.5
U3 - Beta(1, 1)	19.5	6.2	0.9	0.1	0.1	73.2
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	3.5	2.1	0.9	0.4	-
U2 - Beta(0.5, 0.5)	11.1	3.5	2.1	0.9	0.4	-
U2 - Beta(0.7, 0.3)	11	3.7	2.2	0.9	0.3	-
U2 - Beta(1, 1)	11	3.7	2.1	0.9	0.4	-
U3 - Beta(0.3, 0.7)	11	3.6	2.1	1	0.4	-
U3 - Beta(0.5, 0.5)	11.1	3.6	2.2	0.9	0.3	-
U3 - Beta(0.7, 0.3)	11.1	3.6	2.1	0.9	0.3	-
U3 - Beta(1, 1)	11.1	3.6	2.1	0.9	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	1.7	0.7	0.4	0.1	-
U2 - Beta(0.5, 0.5)	3.2	1.8	0.7	0.4	0.2	-
U2 - Beta(0.7, 0.3)	3.2	1.8	0.8	0.4	0.1	-
U2 - Beta(1, 1)	3.2	1.8	0.7	0.4	0.1	-
U3 - Beta(0.3, 0.7)	3.2	1.8	0.7	0.4	0.1	-
U3 - Beta(0.5, 0.5)	3.2	1.7	0.8	0.4	0.1	-
U3 - Beta(0.7, 0.3)	3.2	1.8	0.7	0.4	0.1	-
U3 - Beta(1, 1)	3.2	1.8	0.7	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						643.4
U2 - Beta(0.5, 0.5)						643.7
U2 - Beta(0.7, 0.3)						644.1
U2 - Beta(1, 1)						644.8
U3 - Beta(0.3, 0.7)						642.9
U3 - Beta(0.5, 0.5)						644.9
U3 - Beta(0.7, 0.3)						644.2
U3 - Beta(1, 1)						643.2
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2		0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	29.9	15.1	33.9	12	4.1	5
U2 - Beta(0.5, 0.5)	31.4	15.4	32	12.4	3.7	5.1
U2 - Beta(0.7, 0.3)	32.4	15	33.3	10.5	3.7	5.1
U2 - Beta(1, 1)	31.8	15.2	31.9	11.9	4.1	5.1
U3 - Beta(0.3, 0.7)	34.9	14.8	32	10.4	2.8	5.1
U3 - Beta(0.5, 0.5)	34.4	16.8	29.8	11.5	2.5	5
U3 - Beta(0.7, 0.3)	36.4	16.1	30.4	9.6	2.4	5.1
U3 - Beta(1, 1)	36.6	15.5	29.1	11.1	2.6	5.1
Mean administrations						
U2 - Beta(0.3, 0.7)	10.7	5.1	7.2	4	2.1	-
U2 - Beta(0.5, 0.5)	10.9	5.2	7.2	3.8	2.1	-
U2 - Beta(0.7, 0.3)	10.9	5.1	7.2	3.9	2.1	-
U2 - Beta(1, 1)	10.9	5	7.2	3.9	2.2	-
U3 - Beta(0.3, 0.7)	10.8	5	7.3	3.8	2.1	-
U3 - Beta(0.5, 0.5)	10.8	5.1	7.2	3.9	2.2	-
U3 - Beta(0.7, 0.3)	10.9	5	7.3	3.8	2.1	-
U3 - Beta(1, 1)	10.9	5	7.2	3.9	2.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	2.4	1.9	1.1	0.5	-
U2 - Beta(0.5, 0.5)	3.1	2.5	1.9	1	0.5	-
U2 - Beta(0.7, 0.3)	3.1	2.5	1.9	1	0.5	-
U2 - Beta(1, 1)	3.1	2.4	1.9	1	0.5	-
U3 - Beta(0.3, 0.7)	3.1	2.4	2	1	0.6	-
U3 - Beta(0.5, 0.5)	3.1	2.4	1.9	1.1	0.6	-
U3 - Beta(0.7, 0.3)	3.1	2.4	1.9	1	0.5	-
U3 - Beta(1, 1)	3.1	2.4	1.9	1	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1108
U2 - Beta(0.5, 0.5)						1107
U2 - Beta(0.7, 0.3)						1107.8
U2 - Beta(1, 1)						1108.2
U3 - Beta(0.3, 0.7)						1108.2
U3 - Beta(0.5, 0.5)						1107.5
U3 - Beta(0.7, 0.3)						1108.1
U3 - Beta(1, 1)						1107.9

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.1	38	23.4	7.6	2.7	18.2
U2 - Beta(0.5, 0.5)	10.2	39.2	22.2	7.9	2.2	18.3
U2 - Beta(0.7, 0.3)	10.7	39.7	21.5	7.1	2.6	18.4
U2 - Beta(1, 1)	10.7	38.7	21.9	8	2.4	18.3
U3 - Beta(0.3, 0.7)	11.7	40.5	20.7	7.1	1.8	18.2
U3 - Beta(0.5, 0.5)	12	41.6	19.1	7.1	2	18.2
U3 - Beta(0.7, 0.3)	13.4	41.5	18.1	7.1	1.7	18.2
U3 - Beta(1, 1)	13.8	42.4	17.6	6.3	1.7	18.2
Mean administrations						
U2 - Beta(0.3, 0.7)	7.8	7.2	6.3	3.7	2	-
U2 - Beta(0.5, 0.5)	7.8	7.5	6.1	3.6	2	-
U2 - Beta(0.7, 0.3)	7.8	7.5	6.1	3.5	2.1	-
U2 - Beta(1, 1)	7.8	7.4	6.2	3.6	2	-
U3 - Beta(0.3, 0.7)	7.8	7.2	6.3	3.7	2	-
U3 - Beta(0.5, 0.5)	7.8	7.5	6.1	3.6	2	-
U3 - Beta(0.7, 0.3)	7.8	7.4	6.2	3.6	2	-
U3 - Beta(1, 1)	7.8	7.5	6	3.6	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	3.2	1.7	1.2	0.6	-
U2 - Beta(0.5, 0.5)	2	3.3	1.7	1.1	0.6	-
U2 - Beta(0.7, 0.3)	2	3.2	1.7	1.1	0.6	-
U2 - Beta(1, 1)	2	3.2	1.7	1.1	0.6	-
U3 - Beta(0.3, 0.7)	2	3.2	1.7	1.2	0.6	-
U3 - Beta(0.5, 0.5)	2	3.3	1.7	1.1	0.6	-
U3 - Beta(0.7, 0.3)	2	3.2	1.7	1.1	0.6	-
U3 - Beta(1, 1)	2	3.3	1.6	1.1	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1012
U2 - Beta(0.5, 0.5)						1011.3
U2 - Beta(0.7, 0.3)						1011.2
U2 - Beta(1, 1)						1011.8
U3 - Beta(0.3, 0.7)						1011.1
U3 - Beta(0.5, 0.5)						1011.1
U3 - Beta(0.7, 0.3)						1010.9
U3 - Beta(1, 1)						1011.4
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.3	6.2	22	42.6	21	2.9
U2 - Beta(0.5, 0.5)	6.1	6.5	21.6	42.4	20.6	2.8
U2 - Beta(0.7, 0.3)	7.2	7.5	19.3	44	19.2	2.8
U2 - Beta(1, 1)	7.4	7	17.8	45.1	19.8	2.9
U3 - Beta(0.3, 0.7)	8.7	8.7	18.7	43.1	18	2.8
U3 - Beta(0.5, 0.5)	10.7	8.1	18.7	41.5	18.1	2.9
U3 - Beta(0.7, 0.3)	11.9	8.5	18.9	41.4	16.4	2.9
U3 - Beta(1, 1)	12.8	9.9	16.8	41.3	16.3	2.9
Mean administrations						
U2 - Beta(0.3, 0.7)	4.3	4.1	6.4	7.9	6.7	-
U2 - Beta(0.5, 0.5)	4.3	4.1	6.5	8	6.5	-
U2 - Beta(0.7, 0.3)	4.3	4.2	6.5	8.1	6.3	-
U2 - Beta(1, 1)	4.3	4.3	6.4	8.1	6.4	-
U3 - Beta(0.3, 0.7)	4.2	4.1	6.4	8.1	6.6	-
U3 - Beta(0.5, 0.5)	4.3	4.2	6.5	8	6.5	-
U3 - Beta(0.7, 0.3)	4.3	4.2	6.4	8	6.4	-
U3 - Beta(1, 1)	4.3	4.2	6.3	8.1	6.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	2.7	2	2.6	1.5	-
U2 - Beta(0.5, 0.5)	0.6	2.7	2	2.6	1.5	-
U2 - Beta(0.7, 0.3)	0.6	2.7	2	2.6	1.5	-
U2 - Beta(1, 1)	0.6	2.8	2	2.6	1.5	-
U3 - Beta(0.3, 0.7)	0.6	2.7	2	2.7	1.5	-
U3 - Beta(0.5, 0.5)	0.6	2.7	2	2.6	1.5	-
U3 - Beta(0.7, 0.3)	0.6	2.7	2	2.6	1.5	-
U3 - Beta(1, 1)	0.6	2.7	2	2.6	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1099.6
U2 - Beta(0.5, 0.5)						1097
U2 - Beta(0.7, 0.3)						1095.7
U2 - Beta(1, 1)						1096.5
U3 - Beta(0.3, 0.7)						1100.8
U3 - Beta(0.5, 0.5)						1098.4
U3 - Beta(0.7, 0.3)						1095.5
U3 - Beta(1, 1)						1098.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.6	33.7	25.6	19.4	12.2	5.5
U2 - Beta(0.5, 0.5)	3.9	34.8	25.9	19.3	10.6	5.5
U2 - Beta(0.7, 0.3)	5	35.9	24.2	18.6	10.8	5.5
U2 - Beta(1, 1)	5.5	36.6	25.6	17.8	9	5.5
U3 - Beta(0.3, 0.7)	6.4	41.3	23.9	16	6.9	5.5
U3 - Beta(0.5, 0.5)	7.6	41.7	23.8	15	6.4	5.5
U3 - Beta(0.7, 0.3)	9.7	41.1	22.4	15.5	5.8	5.5
U3 - Beta(1, 1)	11.6	41.2	22.8	13.8	5.1	5.5
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	6.3	6.6	5.8	5.8	-
U2 - Beta(0.5, 0.5)	4.5	6.4	6.6	5.8	5.7	-
U2 - Beta(0.7, 0.3)	4.5	6.5	6.6	5.7	5.6	-
U2 - Beta(1, 1)	4.5	6.5	6.6	5.7	5.6	-
U3 - Beta(0.3, 0.7)	4.5	6.3	6.6	5.8	5.8	-
U3 - Beta(0.5, 0.5)	4.5	6.5	6.6	5.7	5.7	-
U3 - Beta(0.7, 0.3)	4.6	6.5	6.6	5.8	5.5	-
U3 - Beta(1, 1)	4.5	6.4	6.6	5.8	5.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.4	2.1	2.2	2	-
U2 - Beta(0.5, 0.5)	0.7	3.4	2.1	2.2	2	-
U2 - Beta(0.7, 0.3)	0.7	3.5	2.1	2.2	2	-
U2 - Beta(1, 1)	0.7	3.5	2.1	2.2	1.9	-
U3 - Beta(0.3, 0.7)	0.7	3.4	2	2.2	2	-
U3 - Beta(0.5, 0.5)	0.7	3.5	2.1	2.2	2	-
U3 - Beta(0.7, 0.3)	0.7	3.5	2.1	2.2	1.9	-
U3 - Beta(1, 1)	0.7	3.4	2.1	2.2	1.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1020.7
U2 - Beta(0.5, 0.5)						1021.1
U2 - Beta(0.7, 0.3)						1020.3
U2 - Beta(1, 1)						1021
U3 - Beta(0.3, 0.7)						1021.7
U3 - Beta(0.5, 0.5)						1020.6
U3 - Beta(0.7, 0.3)						1021
U3 - Beta(1, 1)						1020.9
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	65.9	14	3.3	1.7	0.5	14.6
U2 - Beta(0.5, 0.5)	65.7	13.6	3.9	1.4	0.4	15
U2 - Beta(0.7, 0.3)	65.1	14.4	3.8	1.3	0.6	14.8
U2 - Beta(1, 1)	66.7	12.6	4.4	1.1	0.5	14.7
U3 - Beta(0.3, 0.7)	67.6	13.3	2.7	1.4	0.4	14.6
U3 - Beta(0.5, 0.5)	67.1	12.8	3.1	1	0.4	15
U3 - Beta(0.7, 0.3)	67.9	12.9	2.7	1.1	0.4	15
U3 - Beta(1, 1)	68.6	12.2	2.9	1.3	0.3	14.7
Mean administrations						
U2 - Beta(0.3, 0.7)	19.2	3.9	2.6	1.1	0.5	-
U2 - Beta(0.5, 0.5)	19.2	4	2.5	1.1	0.4	-
U2 - Beta(0.7, 0.3)	19.2	4	2.5	1.1	0.4	-
U2 - Beta(1, 1)	19.2	3.9	2.5	1.1	0.4	-
U3 - Beta(0.3, 0.7)	19.3	3.8	2.6	1.1	0.5	-
U3 - Beta(0.5, 0.5)	19.2	3.9	2.5	1.1	0.4	-
U3 - Beta(0.7, 0.3)	19.3	4	2.4	1.1	0.4	-
U3 - Beta(1, 1)	19.2	4	2.5	1.1	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.3	1.5	0.8	0.4	0.1	-
U2 - Beta(0.5, 0.5)	5.3	1.6	0.7	0.4	0.1	-
U2 - Beta(0.7, 0.3)	5.3	1.6	0.7	0.4	0.1	-
U2 - Beta(1, 1)	5.3	1.5	0.7	0.4	0.1	-
U3 - Beta(0.3, 0.7)	5.3	1.5	0.7	0.4	0.1	-
U3 - Beta(0.5, 0.5)	5.3	1.5	0.7	0.4	0.1	-
U3 - Beta(0.7, 0.3)	5.3	1.6	0.7	0.4	0.1	-
U3 - Beta(1, 1)	5.3	1.6	0.7	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1059.4
U2 - Beta(0.5, 0.5)						1056.6
U2 - Beta(0.7, 0.3)						1058
U2 - Beta(1, 1)						1058.1
U3 - Beta(0.3, 0.7)						1058.5
U3 - Beta(0.5, 0.5)						1056.2
U3 - Beta(0.7, 0.3)						1056.6
U3 - Beta(1, 1)						1058.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.7	36.5	19.5	11.3	4.2	5.8
U2 - Beta(0.5, 0.5)	23.4	36.4	20	10.4	3.9	5.9
U2 - Beta(0.7, 0.3)	22.8	37.9	19.6	9.3	4.5	5.9
U2 - Beta(1, 1)	22.5	37	20.6	9.6	4.4	5.9
U3 - Beta(0.3, 0.7)	24.7	37.5	18.7	9.6	3.6	5.9
U3 - Beta(0.5, 0.5)	25.6	39.9	17.2	8.8	2.6	5.9
U3 - Beta(0.7, 0.3)	25.4	40.9	17.3	8	2.5	5.9
U3 - Beta(1, 1)	25.5	41.4	16.8	7.8	2.7	5.8
Mean administrations						
U2 - Beta(0.3, 0.7)	10.1	6.9	5.7	3.8	2.6	-
U2 - Beta(0.5, 0.5)	10.1	7	5.6	3.8	2.5	-
U2 - Beta(0.7, 0.3)	10.2	7	5.6	3.8	2.4	-
U2 - Beta(1, 1)	10.1	6.9	5.6	3.8	2.5	-
U3 - Beta(0.3, 0.7)	10.1	6.9	5.6	3.8	2.6	-
U3 - Beta(0.5, 0.5)	10.2	7.1	5.5	3.7	2.5	-
U3 - Beta(0.7, 0.3)	10.2	7	5.6	3.7	2.5	-
U3 - Beta(1, 1)	10.2	7	5.5	3.8	2.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	2.8	1.4	1.1	0.7	-
U2 - Beta(0.5, 0.5)	2.8	2.8	1.4	1.1	0.7	-
U2 - Beta(0.7, 0.3)	2.8	2.8	1.4	1.1	0.6	-
U2 - Beta(1, 1)	2.8	2.8	1.4	1.1	0.7	-
U3 - Beta(0.3, 0.7)	2.8	2.8	1.4	1.1	0.7	-
U3 - Beta(0.5, 0.5)	2.8	2.8	1.3	1.1	0.7	-
U3 - Beta(0.7, 0.3)	2.8	2.8	1.4	1.1	0.7	-
U3 - Beta(1, 1)	2.8	2.8	1.4	1.1	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1119.3
U2 - Beta(0.5, 0.5)						1119.2
U2 - Beta(0.7, 0.3)						1118.5
U2 - Beta(1, 1)						1118.4
U3 - Beta(0.3, 0.7)						1119.2
U3 - Beta(0.5, 0.5)						1118.7
U3 - Beta(0.7, 0.3)						1118.9
U3 - Beta(1, 1)						1119
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	25.4	9.4	3.3	1.3	0.3	60.3
U2 - Beta(0.5, 0.5)	26.2	9.1	3	1	0.3	60.4
U2 - Beta(0.7, 0.3)	27.3	8.3	2.9	1.2	0.3	60
U2 - Beta(1, 1)	26.9	8.8	2.5	1.3	0.2	60.3
U3 - Beta(0.3, 0.7)	26.5	9.5	2.1	1.1	0.2	60.6
U3 - Beta(0.5, 0.5)	27.4	8.4	2.5	1.1	0.3	60.3
U3 - Beta(0.7, 0.3)	28.2	8.8	1.4	1.2	0.2	60.2
U3 - Beta(1, 1)	29.2	7.6	1.8	1.1	0.3	60
Mean administrations						
U2 - Beta(0.3, 0.7)	13.3	3.6	1.9	0.9	0.3	-
U2 - Beta(0.5, 0.5)	13.4	3.6	1.9	0.8	0.3	-
U2 - Beta(0.7, 0.3)	13.4	3.7	1.8	0.8	0.3	-
U2 - Beta(1, 1)	13.4	3.6	1.9	0.8	0.3	-
U3 - Beta(0.3, 0.7)	13.4	3.7	1.9	0.8	0.3	-
U3 - Beta(0.5, 0.5)	13.4	3.6	1.9	0.8	0.3	-
U3 - Beta(0.7, 0.3)	13.4	3.6	1.9	0.8	0.3	-
U3 - Beta(1, 1)	13.4	3.7	1.8	0.8	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.9	1.7	0.6	0.3	0.1	-
U2 - Beta(0.5, 0.5)	3.9	1.6	0.6	0.3	0.1	-
U2 - Beta(0.7, 0.3)	3.9	1.7	0.6	0.3	0.1	-
U2 - Beta(1, 1)	3.9	1.7	0.6	0.3	0.1	-
U3 - Beta(0.3, 0.7)	3.9	1.7	0.6	0.3	0.1	-
U3 - Beta(0.5, 0.5)	3.9	1.7	0.6	0.3	0.1	-
U3 - Beta(0.7, 0.3)	3.9	1.7	0.6	0.3	0.1	-
U3 - Beta(1, 1)	3.9	1.7	0.6	0.3	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						733.1
U2 - Beta(0.5, 0.5)						733.6
U2 - Beta(0.7, 0.3)						733.3
U2 - Beta(1, 1)						733.3
U3 - Beta(0.3, 0.7)						734.9
U3 - Beta(0.5, 0.5)						734.8
U3 - Beta(0.7, 0.3)						733.2
U3 - Beta(1, 1)						734.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.2	8.1	17.9	13	27.6	26.2
U2 - Beta(0.5, 0.5)	8.7	8	17.4	14.6	25.1	26.2
U2 - Beta(0.7, 0.3)	10.3	9.2	16.5	13.4	24.6	26
U2 - Beta(1, 1)	9.9	9.8	15.9	13.4	24.8	26.2
U3 - Beta(0.3, 0.7)	12.5	9	16.7	11.1	24.5	26.2
U3 - Beta(0.5, 0.5)	13.7	10.1	15.3	11.8	22.7	26.4
U3 - Beta(0.7, 0.3)	16.2	10.6	14	11.7	21.2	26.3
U3 - Beta(1, 1)	18.8	11.6	13	10.6	20	26
Mean administrations						
U2 - Beta(0.3, 0.7)	6.8	4.1	5.3	4	5.4	-
U2 - Beta(0.5, 0.5)	6.9	4.1	5.3	4	5.3	-
U2 - Beta(0.7, 0.3)	6.9	4.1	5.3	3.9	5.3	-
U2 - Beta(1, 1)	6.9	4.2	5.2	4	5.2	-
U3 - Beta(0.3, 0.7)	6.9	4	5.3	4	5.4	-
U3 - Beta(0.5, 0.5)	6.9	4.1	5.3	4	5.3	-
U3 - Beta(0.7, 0.3)	6.9	4.3	5.1	4	5.2	-
U3 - Beta(1, 1)	6.9	4.2	5.2	3.9	5.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	2.4	1.8	1.6	1.6	-
U2 - Beta(0.5, 0.5)	1.7	2.4	1.8	1.6	1.5	-
U2 - Beta(0.7, 0.3)	1.7	2.4	1.8	1.6	1.6	-
U2 - Beta(1, 1)	1.7	2.4	1.8	1.6	1.5	-
U3 - Beta(0.3, 0.7)	1.7	2.3	1.8	1.6	1.6	-
U3 - Beta(0.5, 0.5)	1.7	2.4	1.8	1.6	1.5	-
U3 - Beta(0.7, 0.3)	1.7	2.5	1.8	1.6	1.5	-
U3 - Beta(1, 1)	1.7	2.5	1.8	1.6	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						908.9
U2 - Beta(0.5, 0.5)						907.7
U2 - Beta(0.7, 0.3)						907.3
U2 - Beta(1, 1)						907.1
U3 - Beta(0.3, 0.7)						908.9
U3 - Beta(0.5, 0.5)						907
U3 - Beta(0.7, 0.3)						906.8
U3 - Beta(1, 1)						907.8

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6	28.2	12.1	4.8	2.4	46.5
U2 - Beta(0.5, 0.5)	6.4	27.6	13	4.4	2.3	46.3
U2 - Beta(0.7, 0.3)	6.1	29.4	11.2	4.5	2.5	46.3
U2 - Beta(1, 1)	6	29.4	11.5	4.3	2.3	46.5
U3 - Beta(0.3, 0.7)	6.6	29.9	11.1	4.4	1.7	46.3
U3 - Beta(0.5, 0.5)	7.6	29.8	10.6	3.9	1.8	46.3
U3 - Beta(0.7, 0.3)	6.7	32.1	9.7	3.5	1.6	46.4
U3 - Beta(1, 1)	6.9	30.8	10.6	3.8	1.4	46.5
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	5.7	4.1	2.2	1.2	-
U2 - Beta(0.5, 0.5)	8.4	5.8	4.1	2.2	1.2	-
U2 - Beta(0.7, 0.3)	8.4	5.9	4	2.2	1.2	-
U2 - Beta(1, 1)	8.4	5.9	4	2.2	1.2	-
U3 - Beta(0.3, 0.7)	8.4	5.7	4.1	2.2	1.2	-
U3 - Beta(0.5, 0.5)	8.4	5.8	4.1	2.1	1.2	-
U3 - Beta(0.7, 0.3)	8.4	5.9	4	2.2	1.2	-
U3 - Beta(1, 1)	8.4	5.9	4	2.2	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	2.5	1.1	0.7	0.3	-
U2 - Beta(0.5, 0.5)	2.2	2.5	1.1	0.7	0.3	-
U2 - Beta(0.7, 0.3)	2.2	2.5	1.1	0.7	0.3	-
U2 - Beta(1, 1)	2.2	2.5	1.1	0.7	0.3	-
U3 - Beta(0.3, 0.7)	2.2	2.5	1.1	0.7	0.3	-
U3 - Beta(0.5, 0.5)	2.2	2.5	1.1	0.6	0.3	-
U3 - Beta(0.7, 0.3)	2.2	2.5	1.1	0.6	0.3	-
U3 - Beta(1, 1)	2.2	2.5	1.1	0.7	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						814.5
U2 - Beta(0.5, 0.5)						815.1
U2 - Beta(0.7, 0.3)						815.1
U2 - Beta(1, 1)						815.2
U3 - Beta(0.3, 0.7)						815.1
U3 - Beta(0.5, 0.5)						816
U3 - Beta(0.7, 0.3)						815.9
U3 - Beta(1, 1)						815.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	59.3	19.8	4.9	0.6	0.1	15.3
U2 - Beta(0.5, 0.5)	57.9	20.6	5.5	0.8	0	15.2
U2 - Beta(0.7, 0.3)	58.4	20.9	4.4	1	0	15.3
U2 - Beta(1, 1)	58	22	4.2	0.7	0	15.1
U3 - Beta(0.3, 0.7)	61.1	19.3	3.8	0.4	0	15.4
U3 - Beta(0.5, 0.5)	62.1	18.5	3.3	0.7	0	15.4
U3 - Beta(0.7, 0.3)	62.3	19.1	2.5	0.7	0	15.4
U3 - Beta(1, 1)	62.5	18.4	2.9	0.7	0	15.5
Mean administrations						
U2 - Beta(0.3, 0.7)	16.9	5.7	3.6	1.1	0.2	-
U2 - Beta(0.5, 0.5)	17	5.7	3.5	1.1	0.2	-
U2 - Beta(0.7, 0.3)	17	5.8	3.4	1.1	0.2	-
U2 - Beta(1, 1)	16.9	5.8	3.5	1.1	0.2	-
U3 - Beta(0.3, 0.7)	17	5.6	3.6	1	0.2	-
U3 - Beta(0.5, 0.5)	16.9	5.8	3.6	1	0.2	-
U3 - Beta(0.7, 0.3)	16.9	5.8	3.5	1.1	0.2	-
U3 - Beta(1, 1)	17	5.8	3.4	1.1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.2	2.5	1.1	0.4	0.1	-
U2 - Beta(0.5, 0.5)	5.2	2.5	1.1	0.4	0.1	-
U2 - Beta(0.7, 0.3)	5.2	2.5	1.1	0.4	0.1	-
U2 - Beta(1, 1)	5.2	2.5	1.1	0.4	0.1	-
U3 - Beta(0.3, 0.7)	5.2	2.5	1.1	0.4	0.1	-
U3 - Beta(0.5, 0.5)	5.1	2.5	1.1	0.4	0.1	-
U3 - Beta(0.7, 0.3)	5.2	2.5	1.1	0.4	0.1	-
U3 - Beta(1, 1)	5.2	2.5	1.1	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1002.9
U2 - Beta(0.5, 0.5)						1001.8
U2 - Beta(0.7, 0.3)						1002.3
U2 - Beta(1, 1)						1002.3
U3 - Beta(0.3, 0.7)						1002.4
U3 - Beta(0.5, 0.5)						1002.4
U3 - Beta(0.7, 0.3)						1002.7
U3 - Beta(1, 1)						1001.9
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.6	11.3	10	36.6	25	5.5
U2 - Beta(0.5, 0.5)	12	11.6	10.1	36.7	24.3	5.3
U2 - Beta(0.7, 0.3)	12.8	12.4	9	36.1	24.4	5.3
U2 - Beta(1, 1)	13.3	12.4	8.6	34.7	25.5	5.5
U3 - Beta(0.3, 0.7)	16	12.3	8.3	34.9	23	5.5
U3 - Beta(0.5, 0.5)	19.3	11.6	6.8	34.5	22.5	5.3
U3 - Beta(0.7, 0.3)	19.1	12.2	7	34.6	21.6	5.5
U3 - Beta(1, 1)	22.4	11.4	6.3	33	21.6	5.3
Mean administrations						
U2 - Beta(0.3, 0.7)	4.7	4.5	5.2	7.3	7.5	-
U2 - Beta(0.5, 0.5)	4.7	4.5	5.2	7.3	7.4	-
U2 - Beta(0.7, 0.3)	4.8	4.7	5.1	7.3	7.3	-
U2 - Beta(1, 1)	4.7	4.7	5.1	7.3	7.3	-
U3 - Beta(0.3, 0.7)	4.6	4.5	5.2	7.3	7.5	-
U3 - Beta(0.5, 0.5)	4.7	4.5	5.2	7.4	7.3	-
U3 - Beta(0.7, 0.3)	4.7	4.6	5.2	7.3	7.3	-
U3 - Beta(1, 1)	4.8	4.6	5.1	7.3	7.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	2.8	1.7	2.7	1.7	-
U2 - Beta(0.5, 0.5)	0.8	2.9	1.6	2.7	1.6	-
U2 - Beta(0.7, 0.3)	0.8	2.9	1.6	2.7	1.6	-
U2 - Beta(1, 1)	0.8	2.9	1.6	2.7	1.6	-
U3 - Beta(0.3, 0.7)	0.7	2.8	1.7	2.7	1.7	-
U3 - Beta(0.5, 0.5)	0.8	2.8	1.7	2.7	1.6	-
U3 - Beta(0.7, 0.3)	0.8	2.9	1.6	2.7	1.6	-
U3 - Beta(1, 1)	0.8	2.9	1.6	2.7	1.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1077.6
U2 - Beta(0.5, 0.5)						1076.2
U2 - Beta(0.7, 0.3)						1074.4
U2 - Beta(1, 1)						1072.3
U3 - Beta(0.3, 0.7)						1076.4
U3 - Beta(0.5, 0.5)						1075.7
U3 - Beta(0.7, 0.3)						1073.1
U3 - Beta(1, 1)						1075.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.7	30.1	14.5	4.8	1.7	41.2
U2 - Beta(0.5, 0.5)	8.3	29.3	14.3	4.7	2	41.4
U2 - Beta(0.7, 0.3)	8.3	29.9	13.5	4.6	2.2	41.5
U2 - Beta(1, 1)	8.8	30	13.2	5.3	1.5	41.2
U3 - Beta(0.3, 0.7)	8.7	31.5	12	4.7	1.5	41.6
U3 - Beta(0.5, 0.5)	9.8	30.7	12.6	3.8	1.5	41.6
U3 - Beta(0.7, 0.3)	9.3	30.7	13	4.1	1.4	41.5
U3 - Beta(1, 1)	11	31.4	11.3	4.1	1	41.2
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	6.1	4.5	2.4	1.3	-
U2 - Beta(0.5, 0.5)	8.2	6.2	4.4	2.4	1.3	-
U2 - Beta(0.7, 0.3)	8.2	6.2	4.5	2.3	1.2	-
U2 - Beta(1, 1)	8.2	6.2	4.4	2.4	1.2	-
U3 - Beta(0.3, 0.7)	8.2	6.1	4.6	2.4	1.2	-
U3 - Beta(0.5, 0.5)	8.2	6.1	4.5	2.4	1.2	-
U3 - Beta(0.7, 0.3)	8.3	6.2	4.5	2.4	1.2	-
U3 - Beta(1, 1)	8.3	6.1	4.4	2.4	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	2.8	1.4	0.8	0.4	-
U2 - Beta(0.5, 0.5)	2.1	2.8	1.3	0.8	0.4	-
U2 - Beta(0.7, 0.3)	2.2	2.8	1.3	0.8	0.4	-
U2 - Beta(1, 1)	2.2	2.8	1.3	0.8	0.4	-
U3 - Beta(0.3, 0.7)	2.1	2.8	1.4	0.8	0.4	-
U3 - Beta(0.5, 0.5)	2.1	2.8	1.4	0.8	0.4	-
U3 - Beta(0.7, 0.3)	2.2	2.8	1.4	0.8	0.4	-
U3 - Beta(1, 1)	2.2	2.8	1.3	0.8	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						823.7
U2 - Beta(0.5, 0.5)						824.2
U2 - Beta(0.7, 0.3)						822.9
U2 - Beta(1, 1)						823.3
U3 - Beta(0.3, 0.7)						823.5
U3 - Beta(0.5, 0.5)						822.3
U3 - Beta(0.7, 0.3)						824.4
U3 - Beta(1, 1)						823.2
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.8	12.6	14.9	28.1	17.1	12.5
U2 - Beta(0.5, 0.5)	15.7	12.6	13.3	28.8	17.3	12.3
U2 - Beta(0.7, 0.3)	18.5	12.7	13.2	26.5	16.3	12.8
U2 - Beta(1, 1)	19.2	13.9	12.9	26.1	15.3	12.6
U3 - Beta(0.3, 0.7)	20.6	13.8	12.6	25.3	14.9	12.8
U3 - Beta(0.5, 0.5)	24.3	15.2	9.5	25.4	13.1	12.5
U3 - Beta(0.7, 0.3)	28.1	14.5	9.5	23.1	12.2	12.6
U3 - Beta(1, 1)	31.6	15.4	8.5	21.3	10.3	12.9
Mean administrations						
U2 - Beta(0.3, 0.7)	5.8	5	5.8	6	5.4	-
U2 - Beta(0.5, 0.5)	5.9	5	5.8	6.2	5.2	-
U2 - Beta(0.7, 0.3)	6	5.3	5.6	6.1	5.1	-
U2 - Beta(1, 1)	6	5.3	5.4	6.1	5.2	-
U3 - Beta(0.3, 0.7)	5.8	5	5.8	6	5.4	-
U3 - Beta(0.5, 0.5)	5.9	5.1	5.7	6.2	5.2	-
U3 - Beta(0.7, 0.3)	6	5.2	5.6	6.2	5.1	-
U3 - Beta(1, 1)	6	5.3	5.5	6.1	5.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.3	2.9	2	2.4	1.6	-
U2 - Beta(0.5, 0.5)	1.3	2.9	2	2.4	1.5	-
U2 - Beta(0.7, 0.3)	1.3	3	1.9	2.4	1.5	-
U2 - Beta(1, 1)	1.4	3.1	1.9	2.4	1.6	-
U3 - Beta(0.3, 0.7)	1.3	3	2	2.3	1.6	-
U3 - Beta(0.5, 0.5)	1.3	3	2	2.4	1.5	-
U3 - Beta(0.7, 0.3)	1.3	3	1.9	2.4	1.5	-
U3 - Beta(1, 1)	1.3	3.1	1.9	2.4	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						986.3
U2 - Beta(0.5, 0.5)						985.5
U2 - Beta(0.7, 0.3)						981.3
U2 - Beta(1, 1)						983.1
U3 - Beta(0.3, 0.7)						985.2
U3 - Beta(0.5, 0.5)						984.3
U3 - Beta(0.7, 0.3)						982.1
U3 - Beta(1, 1)						982.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	40.3	23.3	13.5	3.7	1	18.2
U2 - Beta(0.5, 0.5)	44.1	22.9	10.4	3.7	0.7	18.2
U2 - Beta(0.7, 0.3)	43.7	23	10.6	3.5	0.9	18.3
U2 - Beta(1, 1)	45.1	22.9	10.6	2.7	0.7	18
U3 - Beta(0.3, 0.7)	51	21.4	7.4	1.8	0.3	18.1
U3 - Beta(0.5, 0.5)	53.4	20.1	6.7	1.3	0.5	18
U3 - Beta(0.7, 0.3)	53.2	20.6	5.9	1.4	0.4	18.5
U3 - Beta(1, 1)	58.2	17	5.4	1.1	0.3	18
Mean administrations						
U2 - Beta(0.3, 0.7)	10	6.4	6	3.3	1.7	-
U2 - Beta(0.5, 0.5)	10	6.6	5.8	3.3	1.7	-
U2 - Beta(0.7, 0.3)	10	6.8	5.7	3.2	1.7	-
U2 - Beta(1, 1)	10.2	6.7	5.7	3.2	1.7	-
U3 - Beta(0.3, 0.7)	10	6.4	6	3.3	1.7	-
U3 - Beta(0.5, 0.5)	10.1	6.6	5.8	3.3	1.7	-
U3 - Beta(0.7, 0.3)	10.1	6.7	5.7	3.3	1.7	-
U3 - Beta(1, 1)	10.1	6.7	5.7	3.2	1.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	3.3	2.1	1.4	0.7	-
U2 - Beta(0.5, 0.5)	3	3.4	2	1.4	0.7	-
U2 - Beta(0.7, 0.3)	3	3.5	2	1.4	0.6	-
U2 - Beta(1, 1)	3	3.4	2	1.4	0.6	-
U3 - Beta(0.3, 0.7)	3	3.3	2	1.4	0.7	-
U3 - Beta(0.5, 0.5)	3	3.4	2	1.4	0.6	-
U3 - Beta(0.7, 0.3)	3	3.4	2	1.4	0.6	-
U3 - Beta(1, 1)	3	3.4	2	1.4	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						933.7
U2 - Beta(0.5, 0.5)						933.7
U2 - Beta(0.7, 0.3)						932.8
U2 - Beta(1, 1)						933.3
U3 - Beta(0.3, 0.7)						934.2
U3 - Beta(0.5, 0.5)						932.7
U3 - Beta(0.7, 0.3)						932
U3 - Beta(1, 1)						933.2
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	29.9	22.9	16.4	17.4	13.3	0.1
U2 - Beta(0.5, 0.5)	30.4	23.1	18.8	15.1	12.5	0.1
U2 - Beta(0.7, 0.3)	32.5	23	18	14.8	11.6	0.1
U2 - Beta(1, 1)	32.8	27.3	15.7	14.2	9.9	0.1
U3 - Beta(0.3, 0.7)	36.4	26.9	15.8	13.5	7.3	0.1
U3 - Beta(0.5, 0.5)	39.5	26.6	15	12.5	6.3	0.1
U3 - Beta(0.7, 0.3)	42.1	25.4	15.2	11.3	5.9	0.1
U3 - Beta(1, 1)	44.9	26.6	13.2	10.2	5	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	6.1	5.3	5.4	5.7	7.5	-
U2 - Beta(0.5, 0.5)	6	5.4	5.5	5.7	7.3	-
U2 - Beta(0.7, 0.3)	6	5.5	5.5	5.9	7.2	-
U2 - Beta(1, 1)	6.1	5.4	5.5	5.8	7.2	-
U3 - Beta(0.3, 0.7)	6	5.3	5.4	5.7	7.6	-
U3 - Beta(0.5, 0.5)	6	5.5	5.5	5.8	7.3	-
U3 - Beta(0.7, 0.3)	6	5.5	5.5	5.8	7.2	-
U3 - Beta(1, 1)	6	5.5	5.5	5.8	7.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.3	3	1.9	2.6	2.9	-
U2 - Beta(0.5, 0.5)	1.3	3.1	1.9	2.6	2.8	-
U2 - Beta(0.7, 0.3)	1.3	3.1	1.9	2.6	2.7	-
U2 - Beta(1, 1)	1.3	3.1	1.9	2.6	2.7	-
U3 - Beta(0.3, 0.7)	1.3	3	1.8	2.5	2.9	-
U3 - Beta(0.5, 0.5)	1.3	3.1	1.9	2.6	2.8	-
U3 - Beta(0.7, 0.3)	1.2	3.2	1.9	2.6	2.7	-
U3 - Beta(1, 1)	1.3	3.1	1.9	2.6	2.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1009.2
U2 - Beta(0.5, 0.5)						1008.8
U2 - Beta(0.7, 0.3)						1008.1
U2 - Beta(1, 1)						1009.2
U3 - Beta(0.3, 0.7)						1009.8
U3 - Beta(0.5, 0.5)						1007.9
U3 - Beta(0.7, 0.3)						1008.1
U3 - Beta(1, 1)						1008.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	30	48.5	17.9	2.5	0.1	1
U2 - Beta(0.5, 0.5)	30.5	48.5	17.7	2.3	0	1
U2 - Beta(0.7, 0.3)	29.2	49.4	18.3	2.1	0	1
U2 - Beta(1, 1)	30.4	49.8	16.5	2.3	0	1
U3 - Beta(0.3, 0.7)	33.7	48.4	15.4	1.4	0.1	1
U3 - Beta(0.5, 0.5)	34.1	49.6	13.8	1.5	0.1	0.9
U3 - Beta(0.7, 0.3)	34	49.6	13.7	1.7	0	1
U3 - Beta(1, 1)	34.2	50.1	13.2	1.5	0	1
Mean administrations						
U2 - Beta(0.3, 0.7)	10	9.7	7.4	2.4	0.4	-
U2 - Beta(0.5, 0.5)	10.1	9.7	7.3	2.4	0.4	-
U2 - Beta(0.7, 0.3)	10.1	9.7	7.2	2.3	0.4	-
U2 - Beta(1, 1)	10	9.8	7.2	2.4	0.4	-
U3 - Beta(0.3, 0.7)	10	9.6	7.4	2.4	0.4	-
U3 - Beta(0.5, 0.5)	10.1	9.6	7.3	2.4	0.4	-
U3 - Beta(0.7, 0.3)	10.2	9.7	7.1	2.3	0.5	-
U3 - Beta(1, 1)	10.1	9.8	7.1	2.4	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	3.4	1.6	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.5	3.4	1.6	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.4	1.6	0.6	0.1	-
U2 - Beta(1, 1)	2.5	3.4	1.6	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.5	3.3	1.7	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.5	3.4	1.6	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.5	3.4	1.6	0.6	0.1	-
U3 - Beta(1, 1)	2.5	3.4	1.6	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1192.7
U2 - Beta(0.5, 0.5)						1191.8
U2 - Beta(0.7, 0.3)						1190.1
U2 - Beta(1, 1)						1191.3
U3 - Beta(0.3, 0.7)						1192.2
U3 - Beta(0.5, 0.5)						1192.1
U3 - Beta(0.7, 0.3)						1190.6
U3 - Beta(1, 1)						1191.4
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.1	8.4	9.3	14.6	59.7	0.9
U2 - Beta(0.5, 0.5)	8.7	8.6	9.2	13.8	58.8	0.9
U2 - Beta(0.7, 0.3)	10.5	8.9	9	13.4	57.2	1
U2 - Beta(1, 1)	12.2	9.1	8.4	12.6	56.7	1
U3 - Beta(0.3, 0.7)	13.3	10.9	10.3	11.8	52.8	0.9
U3 - Beta(0.5, 0.5)	15.4	11.8	9.9	11.4	50.6	0.9
U3 - Beta(0.7, 0.3)	18.8	11.3	8.8	12.6	47.5	1
U3 - Beta(1, 1)	20.1	12.4	9.1	11	46.5	0.9
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	3.7	4.6	5.5	12.5	-
U2 - Beta(0.5, 0.5)	3.7	3.9	4.6	5.5	12.2	-
U2 - Beta(0.7, 0.3)	3.8	3.9	4.6	5.5	12	-
U2 - Beta(1, 1)	3.8	3.9	4.5	5.5	12.2	-
U3 - Beta(0.3, 0.7)	3.7	3.8	4.5	5.5	12.4	-
U3 - Beta(0.5, 0.5)	3.7	3.9	4.5	5.5	12.2	-
U3 - Beta(0.7, 0.3)	3.7	4	4.6	5.5	12.1	-
U3 - Beta(1, 1)	3.8	3.9	4.6	5.4	12.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.6	1.5	2.5	3.4	-
U2 - Beta(0.5, 0.5)	0.3	2.7	1.5	2.5	3.4	-
U2 - Beta(0.7, 0.3)	0.4	2.7	1.5	2.5	3.3	-
U2 - Beta(1, 1)	0.4	2.7	1.5	2.5	3.4	-
U3 - Beta(0.3, 0.7)	0.3	2.7	1.4	2.5	3.4	-
U3 - Beta(0.5, 0.5)	0.3	2.7	1.5	2.5	3.4	-
U3 - Beta(0.7, 0.3)	0.3	2.7	1.5	2.5	3.4	-
U3 - Beta(1, 1)	0.3	2.7	1.5	2.5	3.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1076.8
U2 - Beta(0.5, 0.5)						1074.4
U2 - Beta(0.7, 0.3)						1071.5
U2 - Beta(1, 1)						1073
U3 - Beta(0.3, 0.7)						1075.7
U3 - Beta(0.5, 0.5)						1073.1
U3 - Beta(0.7, 0.3)						1071.5
U3 - Beta(1, 1)						1072.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15	51.7	17.5	1.3	0	14.5
U2 - Beta(0.5, 0.5)	14.6	51.7	17.4	1.6	0	14.7
U2 - Beta(0.7, 0.3)	16	50.9	16.8	2	0	14.3
U2 - Beta(1, 1)	16.5	50.2	16.7	1.8	0	14.8
U3 - Beta(0.3, 0.7)	17.3	52	15.1	1.3	0	14.3
U3 - Beta(0.5, 0.5)	18.1	50.8	14.9	1.8	0	14.4
U3 - Beta(0.7, 0.3)	20.1	50.1	13.1	1.9	0	14.8
U3 - Beta(1, 1)	21.1	49.6	12.9	1.8	0	14.6
Mean administrations						
U2 - Beta(0.3, 0.7)	7.5	10.1	7.2	2.2	0.4	-
U2 - Beta(0.5, 0.5)	7.6	10.3	6.9	2.3	0.4	-
U2 - Beta(0.7, 0.3)	7.6	10.2	7	2.2	0.4	-
U2 - Beta(1, 1)	7.6	10.2	7	2.2	0.4	-
U3 - Beta(0.3, 0.7)	7.5	10	7.1	2.4	0.4	-
U3 - Beta(0.5, 0.5)	7.6	10.2	7.1	2.2	0.4	-
U3 - Beta(0.7, 0.3)	7.6	10.3	6.9	2.2	0.4	-
U3 - Beta(1, 1)	7.6	10.2	6.9	2.2	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	4.4	2	0.8	0.1	-
U2 - Beta(0.5, 0.5)	1.9	4.4	2	0.8	0.1	-
U2 - Beta(0.7, 0.3)	1.9	4.4	2	0.8	0.1	-
U2 - Beta(1, 1)	1.9	4.4	2	0.8	0.1	-
U3 - Beta(0.3, 0.7)	1.8	4.4	2	0.8	0.1	-
U3 - Beta(0.5, 0.5)	1.9	4.4	2	0.8	0.1	-
U3 - Beta(0.7, 0.3)	1.9	4.4	1.9	0.8	0.1	-
U3 - Beta(1, 1)	1.9	4.4	1.9	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1008.9
U2 - Beta(0.5, 0.5)						1006.9
U2 - Beta(0.7, 0.3)						1006
U2 - Beta(1, 1)						1005.6
U3 - Beta(0.3, 0.7)						1008.9
U3 - Beta(0.5, 0.5)						1006.8
U3 - Beta(0.7, 0.3)						1006.7
U3 - Beta(1, 1)						1005.4
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.1	43.3	29.5	9.1	1.8	6.2
U2 - Beta(0.5, 0.5)	11.2	42.1	28.9	10.2	1.4	6.2
U2 - Beta(0.7, 0.3)	12.5	42.6	27.6	10.1	1.1	6.1
U2 - Beta(1, 1)	13.2	42.7	26.4	10.5	1.1	6.1
U3 - Beta(0.3, 0.7)	16	45.8	23.6	7.3	1.1	6.2
U3 - Beta(0.5, 0.5)	16.3	47.3	22.1	7.8	0.5	6
U3 - Beta(0.7, 0.3)	19.3	44.6	21	8.5	0.5	6.1
U3 - Beta(1, 1)	21.5	44.9	20.3	6.7	0.5	6.1
Mean administrations						
U2 - Beta(0.3, 0.7)	5	8.5	8.4	4.7	2.3	-
U2 - Beta(0.5, 0.5)	5.2	8.6	8.2	4.6	2.2	-
U2 - Beta(0.7, 0.3)	5.2	8.7	8.3	4.5	2.3	-
U2 - Beta(1, 1)	5.2	8.8	8.1	4.6	2.2	-
U3 - Beta(0.3, 0.7)	5	8.5	8.3	4.7	2.3	-
U3 - Beta(0.5, 0.5)	5.2	8.7	8.1	4.6	2.3	-
U3 - Beta(0.7, 0.3)	5.3	8.7	8.1	4.5	2.2	-
U3 - Beta(1, 1)	5.2	8.7	8.1	4.5	2.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	4.2	2.6	1.7	0.8	-
U2 - Beta(0.5, 0.5)	1	4.3	2.5	1.7	0.7	-
U2 - Beta(0.7, 0.3)	1	4.3	2.5	1.7	0.7	-
U2 - Beta(1, 1)	1	4.3	2.5	1.7	0.7	-
U3 - Beta(0.3, 0.7)	0.9	4.2	2.6	1.7	0.8	-
U3 - Beta(0.5, 0.5)	1	4.3	2.5	1.7	0.7	-
U3 - Beta(0.7, 0.3)	1	4.3	2.5	1.7	0.7	-
U3 - Beta(1, 1)	1	4.3	2.5	1.7	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1028.2
U2 - Beta(0.5, 0.5)						1025.4
U2 - Beta(0.7, 0.3)						1027
U2 - Beta(1, 1)						1025.5
U3 - Beta(0.3, 0.7)						1027.2
U3 - Beta(0.5, 0.5)						1025.2
U3 - Beta(0.7, 0.3)						1025.7
U3 - Beta(1, 1)						1026.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	61.7	1.5	0	0	0	36.8
U2 - Beta(0.5, 0.5)	61.4	1.8	0	0	0	36.8
U2 - Beta(0.7, 0.3)	61.2	1.8	0.1	0	0	36.9
U2 - Beta(1, 1)	61.5	1.5	0	0	0	37
U3 - Beta(0.3, 0.7)	62.1	1.4	0	0	0	36.5
U3 - Beta(0.5, 0.5)	62	1.4	0	0	0	36.6
U3 - Beta(0.7, 0.3)	61.8	1.4	0	0	0	36.8
U3 - Beta(1, 1)	61.5	1.6	0	0	0	36.9
Mean administrations						
U2 - Beta(0.3, 0.7)	21.1	2.1	0.6	0	0	-
U2 - Beta(0.5, 0.5)	21.1	2.1	0.6	0	0	-
U2 - Beta(0.7, 0.3)	21.1	2.1	0.6	0	0	-
U2 - Beta(1, 1)	21.1	2.1	0.6	0	0	-
U3 - Beta(0.3, 0.7)	21.1	2.1	0.6	0	0	-
U3 - Beta(0.5, 0.5)	21.1	2.1	0.6	0	0	-
U3 - Beta(0.7, 0.3)	21	2.1	0.6	0	0	-
U3 - Beta(1, 1)	21.1	2.1	0.6	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.3	0.6	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4.3	0.6	0.1	0	0	-
U2 - Beta(0.7, 0.3)	4.3	0.7	0.1	0	0	-
U2 - Beta(1, 1)	4.3	0.6	0.1	0	0	-
U3 - Beta(0.3, 0.7)	4.3	0.6	0.1	0	0	-
U3 - Beta(0.5, 0.5)	4.3	0.6	0.1	0	0	-
U3 - Beta(0.7, 0.3)	4.3	0.6	0.1	0	0	-
U3 - Beta(1, 1)	4.3	0.6	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1038.2
U2 - Beta(0.5, 0.5)						1037.4
U2 - Beta(0.7, 0.3)						1038
U2 - Beta(1, 1)						1040.3
U3 - Beta(0.3, 0.7)						1040.7
U3 - Beta(0.5, 0.5)						1039
U3 - Beta(0.7, 0.3)						1037.8
U3 - Beta(1, 1)						1038
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	23.5	52.3	12.3	1.8	0	10.1
U2 - Beta(0.5, 0.5)	23.5	52.4	12.4	2.1	0	9.6
U2 - Beta(0.7, 0.3)	23.4	51.3	13.2	2.2	0	9.9
U2 - Beta(1, 1)	23.5	51.4	13.7	1.6	0	9.8
U3 - Beta(0.3, 0.7)	26.5	51.5	10.3	1.6	0	10.1
U3 - Beta(0.5, 0.5)	27.4	50.8	10.7	1.6	0	9.5
U3 - Beta(0.7, 0.3)	28.6	49	10.2	1.7	0	10.5
U3 - Beta(1, 1)	32.4	46.6	9.8	1.6	0	9.6
Mean administrations						
U2 - Beta(0.3, 0.7)	7.5	12.1	6.6	1.7	0.3	-
U2 - Beta(0.5, 0.5)	7.6	12.2	6.4	1.7	0.3	-
U2 - Beta(0.7, 0.3)	7.7	12.1	6.4	1.7	0.3	-
U2 - Beta(1, 1)	7.8	12	6.4	1.7	0.3	-
U3 - Beta(0.3, 0.7)	7.5	12.1	6.5	1.8	0.3	-
U3 - Beta(0.5, 0.5)	7.6	12.2	6.5	1.7	0.3	-
U3 - Beta(0.7, 0.3)	7.7	12.1	6.3	1.7	0.3	-
U3 - Beta(1, 1)	7.7	12.1	6.3	1.7	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	5.2	1.9	0.6	0.1	-
U2 - Beta(0.5, 0.5)	1.8	5.2	1.9	0.6	0.1	-
U2 - Beta(0.7, 0.3)	1.9	5.2	1.9	0.6	0.1	-
U2 - Beta(1, 1)	1.9	5.2	1.9	0.6	0.1	-
U3 - Beta(0.3, 0.7)	1.8	5.2	1.9	0.6	0.1	-
U3 - Beta(0.5, 0.5)	1.8	5.3	1.9	0.5	0.1	-
U3 - Beta(0.7, 0.3)	1.9	5.3	1.8	0.5	0.1	-
U3 - Beta(1, 1)	1.9	5.2	1.8	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1024.8
U2 - Beta(0.5, 0.5)						1022.2
U2 - Beta(0.7, 0.3)						1022.1
U2 - Beta(1, 1)						1021.6
U3 - Beta(0.3, 0.7)						1024.8
U3 - Beta(0.5, 0.5)						1024.6
U3 - Beta(0.7, 0.3)						1020.4
U3 - Beta(1, 1)						1021.9

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.5	3.9	0.3	0	0	88.3
U2 - Beta(0.5, 0.5)	7.4	3.5	0.3	0	0	88.8
U2 - Beta(0.7, 0.3)	7.3	3.8	0.4	0	0	88.5
U2 - Beta(1, 1)	7.1	4.2	0.4	0	0	88.3
U3 - Beta(0.3, 0.7)	7.2	3.5	0.3	0	0	89
U3 - Beta(0.5, 0.5)	7.4	3.7	0.4	0	0	88.5
U3 - Beta(0.7, 0.3)	7.4	3.8	0.3	0	0	88.5
U3 - Beta(1, 1)	7.2	4	0.3	0	0	88.5
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	2.5	0.6	0	0	-
U2 - Beta(0.5, 0.5)	11.8	2.5	0.6	0	0	-
U2 - Beta(0.7, 0.3)	11.8	2.5	0.6	0	0	-
U2 - Beta(1, 1)	11.8	2.5	0.6	0	0	-
U3 - Beta(0.3, 0.7)	11.7	2.5	0.6	0	0	-
U3 - Beta(0.5, 0.5)	11.8	2.5	0.6	0	0	-
U3 - Beta(0.7, 0.3)	11.8	2.5	0.6	0	0	-
U3 - Beta(1, 1)	11.8	2.5	0.6	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.3	1.3	0.2	0	0	-
U2 - Beta(0.5, 0.5)	3.3	1.3	0.2	0	0	-
U2 - Beta(0.7, 0.3)	3.3	1.3	0.2	0	0	-
U2 - Beta(1, 1)	3.3	1.3	0.2	0	0	-
U3 - Beta(0.3, 0.7)	3.3	1.3	0.2	0	0	-
U3 - Beta(0.5, 0.5)	3.3	1.3	0.2	0	0	-
U3 - Beta(0.7, 0.3)	3.3	1.3	0.2	0	0	-
U3 - Beta(1, 1)	3.3	1.3	0.2	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						546.8
U2 - Beta(0.5, 0.5)						547.7
U2 - Beta(0.7, 0.3)						547.5
U2 - Beta(1, 1)						547.6
U3 - Beta(0.3, 0.7)						546.9
U3 - Beta(0.5, 0.5)						547.6
U3 - Beta(0.7, 0.3)						548
U3 - Beta(1, 1)						547.3
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.8	1.5	4.5	44.7	48.4	0.1
U2 - Beta(0.5, 0.5)	1	1.3	5	47.2	45.4	0.1
U2 - Beta(0.7, 0.3)	1.6	1.6	6	47.7	43	0.1
U2 - Beta(1, 1)	1.2	1.8	6.2	46.5	44.2	0.1
U3 - Beta(0.3, 0.7)	2.5	2.1	6.1	46.2	43	0.1
U3 - Beta(0.5, 0.5)	2.4	2	7.4	48.9	39.3	0
U3 - Beta(0.7, 0.3)	2.7	3.4	7.5	49	37.3	0.1
U3 - Beta(1, 1)	3.2	2.7	8.1	48.7	37.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.2	3.2	3.8	8.1	11.7	-
U2 - Beta(0.5, 0.5)	3.3	3.2	3.8	8.4	11.3	-
U2 - Beta(0.7, 0.3)	3.3	3.3	3.9	8.4	11.2	-
U2 - Beta(1, 1)	3.3	3.2	3.8	8.4	11.3	-
U3 - Beta(0.3, 0.7)	3.2	3.2	3.8	8.2	11.6	-
U3 - Beta(0.5, 0.5)	3.3	3.3	3.8	8.3	11.3	-
U3 - Beta(0.7, 0.3)	3.3	3.2	3.8	8.5	11.1	-
U3 - Beta(1, 1)	3.3	3.3	3.8	8.4	11.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.1	2.4	1.2	3.2	3.2	-
U2 - Beta(0.5, 0.5)	0.1	2.5	1.2	3.3	3.1	-
U2 - Beta(0.7, 0.3)	0.1	2.5	1.2	3.3	3.1	-
U2 - Beta(1, 1)	0.1	2.5	1.2	3.3	3.1	-
U3 - Beta(0.3, 0.7)	0.1	2.4	1.2	3.2	3.2	-
U3 - Beta(0.5, 0.5)	0.1	2.5	1.2	3.3	3.1	-
U3 - Beta(0.7, 0.3)	0.1	2.4	1.2	3.3	3.1	-
U3 - Beta(1, 1)	0.1	2.5	1.2	3.3	3.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1098.2
U2 - Beta(0.5, 0.5)						1097.6
U2 - Beta(0.7, 0.3)						1095.9
U2 - Beta(1, 1)						1096.8
U3 - Beta(0.3, 0.7)						1099.4
U3 - Beta(0.5, 0.5)						1096.5
U3 - Beta(0.7, 0.3)						1094.8
U3 - Beta(1, 1)						1096.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	32.4	26.7	33.3	5	0	2.6
U2 - Beta(0.5, 0.5)	31.5	27.5	33.5	5.1	0	2.4
U2 - Beta(0.7, 0.3)	31	27.7	33.5	5.3	0	2.5
U2 - Beta(1, 1)	32.2	26.9	32.7	5.6	0	2.6
U3 - Beta(0.3, 0.7)	38.2	27	28.3	4.1	0	2.4
U3 - Beta(0.5, 0.5)	37	28.2	28.3	3.9	0	2.6
U3 - Beta(0.7, 0.3)	36.9	28.9	27.1	4.6	0	2.5
U3 - Beta(1, 1)	38.7	27.4	27.2	4.3	0	2.4
Mean administrations						
U2 - Beta(0.3, 0.7)	9	7.8	9.5	2.8	0.5	-
U2 - Beta(0.5, 0.5)	9.1	8	9.2	2.8	0.5	-
U2 - Beta(0.7, 0.3)	9.2	7.9	9.2	2.8	0.5	-
U2 - Beta(1, 1)	9.3	7.9	9.1	2.8	0.5	-
U3 - Beta(0.3, 0.7)	9	7.7	9.5	2.8	0.6	-
U3 - Beta(0.5, 0.5)	9.1	8	9.2	2.8	0.5	-
U3 - Beta(0.7, 0.3)	9.2	7.9	9.2	2.8	0.5	-
U3 - Beta(1, 1)	9.2	8	9.1	2.8	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	3.5	2.5	0.9	0.1	-
U2 - Beta(0.5, 0.5)	2.4	3.5	2.5	0.9	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.5	2.5	0.9	0.1	-
U2 - Beta(1, 1)	2.5	3.5	2.5	0.9	0.1	-
U3 - Beta(0.3, 0.7)	2.4	3.4	2.6	0.9	0.1	-
U3 - Beta(0.5, 0.5)	2.4	3.5	2.5	0.9	0.1	-
U3 - Beta(0.7, 0.3)	2.5	3.5	2.5	0.9	0.1	-
U3 - Beta(1, 1)	2.4	3.5	2.5	0.9	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1116.4
U2 - Beta(0.5, 0.5)						1113.8
U2 - Beta(0.7, 0.3)						1114
U2 - Beta(1, 1)						1113.2
U3 - Beta(0.3, 0.7)						1117.4
U3 - Beta(0.5, 0.5)						1114.4
U3 - Beta(0.7, 0.3)						1113.9
U3 - Beta(1, 1)						1113.9
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.3	23.4	28.7	6.3	0	20.3
U2 - Beta(0.5, 0.5)	19.7	25	28	6.8	0	20.5
U2 - Beta(0.7, 0.3)	20.8	26.1	25.4	7.4	0.1	20.2
U2 - Beta(1, 1)	19.9	25	27.9	6.6	0.1	20.5
U3 - Beta(0.3, 0.7)	22.7	25.3	27.1	4.7	0	20.2
U3 - Beta(0.5, 0.5)	24	25	24.3	6.3	0	20.4
U3 - Beta(0.7, 0.3)	24.7	25.3	23.7	6	0	20.3
U3 - Beta(1, 1)	25.4	25.1	22.8	6.1	0	20.6
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	6.3	8.1	3	0.5	-
U2 - Beta(0.5, 0.5)	9	6.5	7.9	2.9	0.5	-
U2 - Beta(0.7, 0.3)	8.9	6.6	7.9	2.9	0.5	-
U2 - Beta(1, 1)	9	6.5	7.9	2.9	0.5	-
U3 - Beta(0.3, 0.7)	8.9	6.4	8.1	2.9	0.5	-
U3 - Beta(0.5, 0.5)	9	6.5	8	2.9	0.5	-
U3 - Beta(0.7, 0.3)	9	6.5	7.9	2.9	0.5	-
U3 - Beta(1, 1)	9	6.5	7.8	3	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	3.1	2.3	0.9	0.1	-
U2 - Beta(0.5, 0.5)	2.5	3.2	2.3	0.9	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.2	2.3	0.9	0.1	-
U2 - Beta(1, 1)	2.5	3.2	2.2	0.9	0.1	-
U3 - Beta(0.3, 0.7)	2.5	3.1	2.3	0.9	0.1	-
U3 - Beta(0.5, 0.5)	2.5	3.1	2.3	0.9	0.1	-
U3 - Beta(0.7, 0.3)	2.5	3.2	2.2	0.9	0.1	-
U3 - Beta(1, 1)	2.5	3.2	2.2	0.9	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						985.7
U2 - Beta(0.5, 0.5)						983.3
U2 - Beta(0.7, 0.3)						984.4
U2 - Beta(1, 1)						984.5
U3 - Beta(0.3, 0.7)						985.9
U3 - Beta(0.5, 0.5)						984.6
U3 - Beta(0.7, 0.3)						984.5
U3 - Beta(1, 1)						985.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.7	21.2	42.9	8.9	0.8	11.5
U2 - Beta(0.5, 0.5)	14.8	21.6	41.5	9.5	1	11.6
U2 - Beta(0.7, 0.3)	14.9	22.1	41.2	9.8	0.6	11.4
U2 - Beta(1, 1)	14.9	21.4	41.5	10.1	0.6	11.5
U3 - Beta(0.3, 0.7)	19.7	22.2	38.4	7.9	0.3	11.5
U3 - Beta(0.5, 0.5)	19.3	22.8	37.7	8.3	0.5	11.4
U3 - Beta(0.7, 0.3)	19.2	22.6	36.6	9.4	0.7	11.5
U3 - Beta(1, 1)	20.5	24.3	35	8.2	0.7	11.3
Mean administrations						
U2 - Beta(0.3, 0.7)	6.9	6.2	9.6	4	1.5	-
U2 - Beta(0.5, 0.5)	7	6.4	9.4	4	1.4	-
U2 - Beta(0.7, 0.3)	7	6.4	9.3	4	1.4	-
U2 - Beta(1, 1)	7.1	6.3	9.3	4	1.5	-
U3 - Beta(0.3, 0.7)	7	6.1	9.6	4.1	1.4	-
U3 - Beta(0.5, 0.5)	7	6.2	9.5	4	1.5	-
U3 - Beta(0.7, 0.3)	7.1	6.3	9.4	4	1.4	-
U3 - Beta(1, 1)	7.1	6.4	9.3	4	1.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	3.1	2.7	1.2	0.4	-
U2 - Beta(0.5, 0.5)	1.7	3.1	2.7	1.2	0.4	-
U2 - Beta(0.7, 0.3)	1.7	3.2	2.6	1.2	0.4	-
U2 - Beta(1, 1)	1.8	3.1	2.6	1.2	0.4	-
U3 - Beta(0.3, 0.7)	1.7	3.1	2.7	1.2	0.4	-
U3 - Beta(0.5, 0.5)	1.7	3.1	2.7	1.2	0.4	-
U3 - Beta(0.7, 0.3)	1.8	3.2	2.7	1.2	0.3	-
U3 - Beta(1, 1)	1.7	3.2	2.6	1.2	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1052.6
U2 - Beta(0.5, 0.5)						1051.8
U2 - Beta(0.7, 0.3)						1050.4
U2 - Beta(1, 1)						1051
U3 - Beta(0.3, 0.7)						1054.1
U3 - Beta(0.5, 0.5)						1051.4
U3 - Beta(0.7, 0.3)						1050.7
U3 - Beta(1, 1)						1050.4
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.9	35.9	6.4	0.2	0	43.6
U2 - Beta(0.5, 0.5)	13.9	35.5	6.5	0.5	0	43.6
U2 - Beta(0.7, 0.3)	14.1	35.1	7.3	0.2	0	43.3
U2 - Beta(1, 1)	14.7	34.7	6.7	0.4	0	43.5
U3 - Beta(0.3, 0.7)	15	36.1	5.2	0.3	0	43.4
U3 - Beta(0.5, 0.5)	15.6	35.8	4.9	0.3	0	43.4
U3 - Beta(0.7, 0.3)	16.2	35.3	4.9	0.2	0	43.4
U3 - Beta(1, 1)	17	34.3	5.1	0.2	0	43.4
Mean administrations						
U2 - Beta(0.3, 0.7)	8.7	8.3	4.3	0.9	0.1	-
U2 - Beta(0.5, 0.5)	8.8	8.3	4.3	0.9	0.1	-
U2 - Beta(0.7, 0.3)	8.8	8.4	4.2	0.9	0.1	-
U2 - Beta(1, 1)	8.8	8.3	4.3	0.9	0.1	-
U3 - Beta(0.3, 0.7)	8.7	8.3	4.4	0.9	0.2	-
U3 - Beta(0.5, 0.5)	8.8	8.4	4.2	0.9	0.1	-
U3 - Beta(0.7, 0.3)	8.8	8.3	4.3	0.9	0.1	-
U3 - Beta(1, 1)	8.8	8.3	4.2	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	3.4	1.2	0.3	0	-
U2 - Beta(0.5, 0.5)	2.3	3.4	1.2	0.3	0	-
U2 - Beta(0.7, 0.3)	2.3	3.4	1.2	0.3	0	-
U2 - Beta(1, 1)	2.4	3.4	1.2	0.3	0	-
U3 - Beta(0.3, 0.7)	2.3	3.4	1.2	0.3	0	-
U3 - Beta(0.5, 0.5)	2.3	3.4	1.2	0.3	0	-
U3 - Beta(0.7, 0.3)	2.3	3.4	1.2	0.3	0	-
U3 - Beta(1, 1)	2.4	3.4	1.2	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						829.9
U2 - Beta(0.5, 0.5)						830.2
U2 - Beta(0.7, 0.3)						829.5
U2 - Beta(1, 1)						831.9
U3 - Beta(0.3, 0.7)						832.9
U3 - Beta(0.5, 0.5)						830.2
U3 - Beta(0.7, 0.3)						830.4
U3 - Beta(1, 1)						829.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	67.5	25.2	4.2	0.3	0	2.8
U2 - Beta(0.5, 0.5)	69	23.6	4.4	0.3	0	2.7
U2 - Beta(0.7, 0.3)	68.1	24.7	4.1	0.3	0	2.8
U2 - Beta(1, 1)	68.6	24.5	4	0.3	0	2.6
U3 - Beta(0.3, 0.7)	70	24.1	3	0.3	0	2.6
U3 - Beta(0.5, 0.5)	69.9	23.4	3.8	0.2	0	2.7
U3 - Beta(0.7, 0.3)	70.5	23	3.6	0.3	0	2.6
U3 - Beta(1, 1)	70.6	22.9	3.6	0.2	0	2.7
Mean administrations						
U2 - Beta(0.3, 0.7)	18	7.2	3.6	0.6	0	-
U2 - Beta(0.5, 0.5)	18	7.3	3.5	0.6	0	-
U2 - Beta(0.7, 0.3)	18.1	7.3	3.6	0.5	0	-
U2 - Beta(1, 1)	18.1	7.2	3.6	0.6	0.1	-
U3 - Beta(0.3, 0.7)	17.9	7.3	3.6	0.5	0	-
U3 - Beta(0.5, 0.5)	18.1	7.3	3.5	0.5	0	-
U3 - Beta(0.7, 0.3)	18.1	7.2	3.5	0.5	0.1	-
U3 - Beta(1, 1)	18.1	7.3	3.5	0.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.7	2.4	1	0.2	0	-
U2 - Beta(0.5, 0.5)	4.6	2.4	1	0.2	0	-
U2 - Beta(0.7, 0.3)	4.7	2.4	1	0.2	0	-
U2 - Beta(1, 1)	4.7	2.4	1	0.2	0	-
U3 - Beta(0.3, 0.7)	4.6	2.4	1	0.2	0	-
U3 - Beta(0.5, 0.5)	4.7	2.4	1	0.2	0	-
U3 - Beta(0.7, 0.3)	4.7	2.4	1	0.2	0	-
U3 - Beta(1, 1)	4.7	2.4	0.9	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1177.8
U2 - Beta(0.5, 0.5)						1177.3
U2 - Beta(0.7, 0.3)						1176.6
U2 - Beta(1, 1)						1177.8
U3 - Beta(0.3, 0.7)						1178.9
U3 - Beta(0.5, 0.5)						1177
U3 - Beta(0.7, 0.3)						1177
U3 - Beta(1, 1)						1177.1
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	18.6	14.7	14.4	38	12.6	1.7
U2 - Beta(0.5, 0.5)	19.3	14	15.8	37.2	12	1.7
U2 - Beta(0.7, 0.3)	19.7	14.3	15	35.6	13.5	1.9
U2 - Beta(1, 1)	20.5	14.6	13.9	37.2	12	1.8
U3 - Beta(0.3, 0.7)	24.1	15	13.8	34.5	10.8	1.8
U3 - Beta(0.5, 0.5)	26.7	15.4	12.7	33	10.5	1.7
U3 - Beta(0.7, 0.3)	29.4	15.6	11.7	31.2	10.4	1.7
U3 - Beta(1, 1)	29.6	16.3	11	32	9.4	1.7
Mean administrations						
U2 - Beta(0.3, 0.7)	5	4.4	5.7	8.2	6.5	-
U2 - Beta(0.5, 0.5)	5	4.5	5.8	8	6.4	-
U2 - Beta(0.7, 0.3)	5	4.6	5.8	8	6.3	-
U2 - Beta(1, 1)	5	4.6	5.7	8	6.4	-
U3 - Beta(0.3, 0.7)	5	4.4	5.6	8.1	6.6	-
U3 - Beta(0.5, 0.5)	5	4.5	5.7	8.2	6.3	-
U3 - Beta(0.7, 0.3)	5	4.6	5.8	8	6.3	-
U3 - Beta(1, 1)	5	4.6	5.7	8.1	6.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	2.7	1.8	2.8	1.6	-
U2 - Beta(0.5, 0.5)	0.9	2.7	1.9	2.8	1.6	-
U2 - Beta(0.7, 0.3)	0.9	2.8	1.9	2.8	1.6	-
U2 - Beta(1, 1)	0.9	2.8	1.8	2.8	1.5	-
U3 - Beta(0.3, 0.7)	0.9	2.7	1.8	2.8	1.6	-
U3 - Beta(0.5, 0.5)	0.9	2.7	1.8	2.9	1.5	-
U3 - Beta(0.7, 0.3)	0.9	2.8	1.8	2.8	1.5	-
U3 - Beta(1, 1)	0.9	2.8	1.8	2.8	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1097.8
U2 - Beta(0.5, 0.5)						1096
U2 - Beta(0.7, 0.3)						1095.8
U2 - Beta(1, 1)						1097.6
U3 - Beta(0.3, 0.7)						1100.1
U3 - Beta(0.5, 0.5)						1097.2
U3 - Beta(0.7, 0.3)						1096.3
U3 - Beta(1, 1)						1097.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.1	30.9	28.2	26.8	6.8	0.2
U2 - Beta(0.5, 0.5)	7.4	32.7	27.8	24.1	7.8	0.2
U2 - Beta(0.7, 0.3)	7.4	36.9	25.8	22.3	7.4	0.2
U2 - Beta(1, 1)	7.7	33.3	26.8	24.5	7.5	0.2
U3 - Beta(0.3, 0.7)	10.4	36.5	26.1	22.1	4.7	0.2
U3 - Beta(0.5, 0.5)	9.7	41.8	24	20.5	3.8	0.2
U3 - Beta(0.7, 0.3)	8.9	42.3	24	19.1	5.5	0.2
U3 - Beta(1, 1)	9.8	42.2	24	20.3	3.5	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.1	5.6	6.4	7.6	6.2	-
U2 - Beta(0.5, 0.5)	4.1	5.6	6.5	7.8	6	-
U2 - Beta(0.7, 0.3)	4.2	5.7	6.5	7.6	5.9	-
U2 - Beta(1, 1)	4.1	5.7	6.4	7.7	6	-
U3 - Beta(0.3, 0.7)	4.1	5.6	6.5	7.7	6.1	-
U3 - Beta(0.5, 0.5)	4.1	5.7	6.4	7.7	6.1	-
U3 - Beta(0.7, 0.3)	4.2	5.8	6.4	7.6	6	-
U3 - Beta(1, 1)	4.1	5.7	6.4	7.7	6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	2.8	1.6	2	1.5	-
U2 - Beta(0.5, 0.5)	0.5	2.8	1.6	2	1.5	-
U2 - Beta(0.7, 0.3)	0.5	2.8	1.7	2	1.5	-
U2 - Beta(1, 1)	0.5	2.8	1.6	2	1.5	-
U3 - Beta(0.3, 0.7)	0.5	2.8	1.7	2	1.5	-
U3 - Beta(0.5, 0.5)	0.5	2.8	1.6	2	1.5	-
U3 - Beta(0.7, 0.3)	0.5	2.9	1.6	2	1.5	-
U3 - Beta(1, 1)	0.5	2.9	1.6	2	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1190.8
U2 - Beta(0.5, 0.5)						1189.7
U2 - Beta(0.7, 0.3)						1187.7
U2 - Beta(1, 1)						1189.4
U3 - Beta(0.3, 0.7)						1190.2
U3 - Beta(0.5, 0.5)						1189.2
U3 - Beta(0.7, 0.3)						1187.7
U3 - Beta(1, 1)						1188.8
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.6	14	33.8	26.3	15.4	0.9
U2 - Beta(0.5, 0.5)	8.5	13.8	35.6	26.7	14.5	0.9
U2 - Beta(0.7, 0.3)	9.9	14.8	35.4	26.1	12.9	0.9
U2 - Beta(1, 1)	11	14.7	34.1	26.8	12.5	0.9
U3 - Beta(0.3, 0.7)	14.3	17.4	34.2	23.9	9.2	1
U3 - Beta(0.5, 0.5)	14.5	16.6	34.7	24.1	9.2	0.9
U3 - Beta(0.7, 0.3)	16	18.2	33.6	23.2	8.1	0.9
U3 - Beta(1, 1)	19.9	18.4	32.4	21.5	6.9	0.9
Mean administrations						
U2 - Beta(0.3, 0.7)	3.9	4.2	7.6	7	7.2	-
U2 - Beta(0.5, 0.5)	4	4.3	7.7	7.1	6.9	-
U2 - Beta(0.7, 0.3)	4	4.4	7.6	7	6.8	-
U2 - Beta(1, 1)	4	4.3	7.7	7	6.9	-
U3 - Beta(0.3, 0.7)	3.9	4.2	7.5	7.1	7.2	-
U3 - Beta(0.5, 0.5)	4	4.3	7.6	7	6.9	-
U3 - Beta(0.7, 0.3)	4	4.4	7.7	7	6.8	-
U3 - Beta(1, 1)	4	4.3	7.6	7	6.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	2.4	2.7	2.4	-
U2 - Beta(0.5, 0.5)	0.4	2.8	2.4	2.7	2.3	-
U2 - Beta(0.7, 0.3)	0.5	2.9	2.4	2.7	2.3	-
U2 - Beta(1, 1)	0.4	2.8	2.5	2.7	2.3	-
U3 - Beta(0.3, 0.7)	0.4	2.8	2.4	2.7	2.4	-
U3 - Beta(0.5, 0.5)	0.4	2.8	2.4	2.7	2.3	-
U3 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.7	2.2	-
U3 - Beta(1, 1)	0.4	2.8	2.4	2.7	2.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1058.9
U2 - Beta(0.5, 0.5)						1055.6
U2 - Beta(0.7, 0.3)						1054.6
U2 - Beta(1, 1)						1055.7
U3 - Beta(0.3, 0.7)						1058.9
U3 - Beta(0.5, 0.5)						1055.9
U3 - Beta(0.7, 0.3)						1056.1
U3 - Beta(1, 1)						1056.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.3	4.9	13.5	38.5	35.6	0.2
U2 - Beta(0.5, 0.5)	8.4	4.8	11.8	40.9	33.8	0.3
U2 - Beta(0.7, 0.3)	8.8	6	13.2	39.7	32.2	0.1
U2 - Beta(1, 1)	9.1	5.5	12.2	40.3	32.8	0.1
U3 - Beta(0.3, 0.7)	10.3	6.4	14.6	39.3	29.3	0.1
U3 - Beta(0.5, 0.5)	13.1	6	13.2	40.1	27.3	0.3
U3 - Beta(0.7, 0.3)	14.6	7.5	13.3	38.5	26	0.1
U3 - Beta(1, 1)	17.1	7.5	12.2	38	25.1	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.4	3.6	5.2	7.8	10	-
U2 - Beta(0.5, 0.5)	3.5	3.6	5.2	7.9	9.7	-
U2 - Beta(0.7, 0.3)	3.5	3.7	5.2	8.1	9.5	-
U2 - Beta(1, 1)	3.5	3.6	5.1	8.1	9.6	-
U3 - Beta(0.3, 0.7)	3.5	3.5	5.2	7.9	10	-
U3 - Beta(0.5, 0.5)	3.5	3.6	5.2	8	9.7	-
U3 - Beta(0.7, 0.3)	3.5	3.7	5.2	8	9.6	-
U3 - Beta(1, 1)	3.6	3.7	5.1	8	9.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.2	2.6	1.6	3	2.5	-
U2 - Beta(0.5, 0.5)	0.2	2.6	1.6	3	2.4	-
U2 - Beta(0.7, 0.3)	0.2	2.6	1.6	3.1	2.4	-
U2 - Beta(1, 1)	0.2	2.7	1.6	3.1	2.4	-
U3 - Beta(0.3, 0.7)	0.2	2.6	1.6	3	2.5	-
U3 - Beta(0.5, 0.5)	0.2	2.6	1.6	3	2.4	-
U3 - Beta(0.7, 0.3)	0.2	2.7	1.6	3	2.4	-
U3 - Beta(1, 1)	0.2	2.7	1.6	3.1	2.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1108.7
U2 - Beta(0.5, 0.5)						1106.6
U2 - Beta(0.7, 0.3)						1107.1
U2 - Beta(1, 1)						1106.2
U3 - Beta(0.3, 0.7)						1109.7
U3 - Beta(0.5, 0.5)						1106.4
U3 - Beta(0.7, 0.3)						1105.5
U3 - Beta(1, 1)						1106.1
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.3	12.6	47.9	17.8	6.4	4
U2 - Beta(0.5, 0.5)	12.7	11.3	47.7	18.4	5.7	4.2
U2 - Beta(0.7, 0.3)	13.1	11.5	45.9	20.6	4.8	4.1
U2 - Beta(1, 1)	13.9	12.9	45.1	19.4	4.6	4.1
U3 - Beta(0.3, 0.7)	16.7	13.6	44.5	16.8	4.5	3.9
U3 - Beta(0.5, 0.5)	18.2	13.1	44.8	16.7	2.9	4.3
U3 - Beta(0.7, 0.3)	22.5	13.2	41.3	15.3	3.5	4.2
U3 - Beta(1, 1)	25.2	13.7	39.1	15	2.9	4.1
Mean administrations						
U2 - Beta(0.3, 0.7)	5	5.4	9.9	5.4	3.7	-
U2 - Beta(0.5, 0.5)	5	5.5	9.8	5.5	3.5	-
U2 - Beta(0.7, 0.3)	5.1	5.6	9.8	5.4	3.5	-
U2 - Beta(1, 1)	5.1	5.5	9.8	5.4	3.5	-
U3 - Beta(0.3, 0.7)	4.9	5.4	10	5.4	3.6	-
U3 - Beta(0.5, 0.5)	5	5.5	9.8	5.3	3.5	-
U3 - Beta(0.7, 0.3)	5.1	5.5	9.8	5.4	3.4	-
U3 - Beta(1, 1)	5.1	5.5	9.8	5.4	3.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.1	3.1	1.9	1.1	-
U2 - Beta(0.5, 0.5)	0.9	3.2	3.1	1.9	1.1	-
U2 - Beta(0.7, 0.3)	0.9	3.2	3	1.9	1.1	-
U2 - Beta(1, 1)	1	3.1	3.1	1.9	1.1	-
U3 - Beta(0.3, 0.7)	0.9	3.1	3.1	1.9	1.1	-
U3 - Beta(0.5, 0.5)	0.9	3.2	3.1	1.9	1.1	-
U3 - Beta(0.7, 0.3)	1	3.2	3.1	1.9	1.1	-
U3 - Beta(1, 1)	0.9	3.2	3.1	1.9	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1053.6
U2 - Beta(0.5, 0.5)						1049
U2 - Beta(0.7, 0.3)						1050.7
U2 - Beta(1, 1)						1049.5
U3 - Beta(0.3, 0.7)						1052.8
U3 - Beta(0.5, 0.5)						1049.2
U3 - Beta(0.7, 0.3)						1048.9
U3 - Beta(1, 1)						1048.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.9	39.2	33.3	11.8	4.6	0.2
U2 - Beta(0.5, 0.5)	12.7	37.5	32.3	12.8	4.5	0.2
U2 - Beta(0.7, 0.3)	12.6	39.1	31.3	12.3	4.5	0.2
U2 - Beta(1, 1)	12.5	39.5	31.7	12.4	3.7	0.2
U3 - Beta(0.3, 0.7)	15.2	43.1	28	10.4	3.1	0.2
U3 - Beta(0.5, 0.5)	17	43	27.5	9.8	2.5	0.2
U3 - Beta(0.7, 0.3)	17.6	43.4	26.9	9.4	2.5	0.2
U3 - Beta(1, 1)	18.2	45.1	25.1	8.9	2.5	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	8.2	9	5	3.2	-
U2 - Beta(0.5, 0.5)	4.6	8.2	9	5	3.2	-
U2 - Beta(0.7, 0.3)	4.7	8.3	8.9	4.9	3.1	-
U2 - Beta(1, 1)	4.6	8.3	8.8	5	3.2	-
U3 - Beta(0.3, 0.7)	4.6	8	9	5.1	3.3	-
U3 - Beta(0.5, 0.5)	4.7	8.2	8.9	5	3.2	-
U3 - Beta(0.7, 0.3)	4.7	8.2	8.9	5	3.1	-
U3 - Beta(1, 1)	4.6	8.3	8.8	5	3.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	3.6	2.2	1.5	0.9	-
U2 - Beta(0.5, 0.5)	0.6	3.6	2.2	1.5	0.8	-
U2 - Beta(0.7, 0.3)	0.7	3.7	2.2	1.4	0.8	-
U2 - Beta(1, 1)	0.7	3.6	2.2	1.5	0.9	-
U3 - Beta(0.3, 0.7)	0.6	3.6	2.2	1.5	0.9	-
U3 - Beta(0.5, 0.5)	0.7	3.6	2.2	1.5	0.9	-
U3 - Beta(0.7, 0.3)	0.7	3.7	2.2	1.5	0.8	-
U3 - Beta(1, 1)	0.7	3.6	2.2	1.5	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1166.5
U2 - Beta(0.5, 0.5)						1166.3
U2 - Beta(0.7, 0.3)						1165.1
U2 - Beta(1, 1)						1166.3
U3 - Beta(0.3, 0.7)						1166.9
U3 - Beta(0.5, 0.5)						1166.2
U3 - Beta(0.7, 0.3)						1165
U3 - Beta(1, 1)						1166.2
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14	36.4	16.2	6.7	2.6	24.1
U2 - Beta(0.5, 0.5)	14.9	36.5	16.1	6.4	1.8	24.3
U2 - Beta(0.7, 0.3)	15	37.8	14.5	6.9	1.9	23.9
U2 - Beta(1, 1)	16.9	37.8	13.3	6.1	1.5	24.4
U3 - Beta(0.3, 0.7)	18.8	36.1	13.5	6.2	1.5	23.9
U3 - Beta(0.5, 0.5)	18.9	37.2	13.2	5.7	1.1	23.9
U3 - Beta(0.7, 0.3)	20.5	38.7	10.5	5.3	1.2	23.8
U3 - Beta(1, 1)	23	36.6	10.2	4.4	1.2	24.6
Mean administrations						
U2 - Beta(0.3, 0.7)	7.6	8.1	5.9	2.8	1.4	-
U2 - Beta(0.5, 0.5)	7.7	8.1	5.7	2.8	1.4	-
U2 - Beta(0.7, 0.3)	7.7	8.2	5.6	2.8	1.4	-
U2 - Beta(1, 1)	7.8	8.1	5.6	2.8	1.4	-
U3 - Beta(0.3, 0.7)	7.6	8	5.8	2.9	1.4	-
U3 - Beta(0.5, 0.5)	7.7	8.1	5.7	2.9	1.4	-
U3 - Beta(0.7, 0.3)	7.8	8.1	5.6	2.8	1.4	-
U3 - Beta(1, 1)	7.8	8.2	5.6	2.8	1.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	3.9	1.9	1	0.4	-
U2 - Beta(0.5, 0.5)	1.9	3.9	1.8	1.1	0.4	-
U2 - Beta(0.7, 0.3)	1.9	3.9	1.8	1	0.4	-
U2 - Beta(1, 1)	2	3.9	1.8	1.1	0.4	-
U3 - Beta(0.3, 0.7)	1.9	3.8	1.9	1.1	0.5	-
U3 - Beta(0.5, 0.5)	1.9	3.8	1.8	1.1	0.4	-
U3 - Beta(0.7, 0.3)	1.9	3.9	1.8	1	0.4	-
U3 - Beta(1, 1)	2	3.9	1.8	1	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						915.8
U2 - Beta(0.5, 0.5)						915.4
U2 - Beta(0.7, 0.3)						914.5
U2 - Beta(1, 1)						915.1
U3 - Beta(0.3, 0.7)						917.1
U3 - Beta(0.5, 0.5)						915.2
U3 - Beta(0.7, 0.3)						914.7
U3 - Beta(1, 1)						915.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	19.2	46.1	14.7	3.1	0.7	16.2
U2 - Beta(0.5, 0.5)	21.3	45.2	13.2	3.2	0.7	16.4
U2 - Beta(0.7, 0.3)	20.9	46.4	13.7	2.5	0.4	16.1
U2 - Beta(1, 1)	22	47.1	11.8	2.4	0.4	16.3
U3 - Beta(0.3, 0.7)	24.1	45.6	11.1	2.6	0.4	16.2
U3 - Beta(0.5, 0.5)	26.6	44.3	10	2.3	0.3	16.5
U3 - Beta(0.7, 0.3)	27.6	44.4	9.3	2.1	0.4	16.2
U3 - Beta(1, 1)	29.4	44.1	8.5	1.8	0.2	16
Mean administrations						
U2 - Beta(0.3, 0.7)	7.1	10.3	6.3	2.4	0.9	-
U2 - Beta(0.5, 0.5)	7.2	10.4	6.2	2.3	1	-
U2 - Beta(0.7, 0.3)	7.3	10.5	6.1	2.3	0.9	-
U2 - Beta(1, 1)	7.3	10.6	5.9	2.3	0.9	-
U3 - Beta(0.3, 0.7)	7.1	10.3	6.3	2.4	0.9	-
U3 - Beta(0.5, 0.5)	7.2	10.4	6.1	2.4	1	-
U3 - Beta(0.7, 0.3)	7.3	10.5	6	2.3	0.9	-
U3 - Beta(1, 1)	7.3	10.5	6	2.3	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.6	4.8	2	0.9	0.3	-
U2 - Beta(0.5, 0.5)	1.7	4.8	1.9	0.9	0.3	-
U2 - Beta(0.7, 0.3)	1.7	4.9	1.9	0.9	0.3	-
U2 - Beta(1, 1)	1.7	4.9	1.8	0.9	0.3	-
U3 - Beta(0.3, 0.7)	1.6	4.8	1.9	1	0.3	-
U3 - Beta(0.5, 0.5)	1.6	4.9	1.9	0.9	0.3	-
U3 - Beta(0.7, 0.3)	1.7	4.9	1.8	0.9	0.3	-
U3 - Beta(1, 1)	1.7	4.9	1.9	0.9	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						955.4
U2 - Beta(0.5, 0.5)						955.3
U2 - Beta(0.7, 0.3)						954.4
U2 - Beta(1, 1)						953.9
U3 - Beta(0.3, 0.7)						957.3
U3 - Beta(0.5, 0.5)						955.3
U3 - Beta(0.7, 0.3)						952.9
U3 - Beta(1, 1)						954.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.9	11.4	3.1	0.9	0.4	70.3
U2 - Beta(0.5, 0.5)	16	12.2	2.5	0.7	0.2	68.4
U2 - Beta(0.7, 0.3)	16.4	11.9	2.7	0.6	0.3	68.1
U2 - Beta(1, 1)	16.3	11.7	2.3	0.7	0.2	68.8
U3 - Beta(0.3, 0.7)	14.5	12.3	1.8	0.4	0.2	70.8
U3 - Beta(0.5, 0.5)	17.2	11	2	0.4	0.1	69.3
U3 - Beta(0.7, 0.3)	15.7	12.1	2.4	0.4	0	69.4
U3 - Beta(1, 1)	15.4	12.1	2.2	0.5	0.1	69.7
Mean administrations						
U2 - Beta(0.3, 0.7)	10.7	5.7	2.6	1.1	0.3	-
U2 - Beta(0.5, 0.5)	10.7	5.7	2.5	1.4	0.4	-
U2 - Beta(0.7, 0.3)	10.7	5.8	2.5	1.4	0.4	-
U2 - Beta(1, 1)	10.7	5.8	2.5	1.4	0.4	-
U3 - Beta(0.3, 0.7)	10.9	5.7	2.4	1.1	0.3	-
U3 - Beta(0.5, 0.5)	10.8	5.7	2.5	1.4	0.4	-
U3 - Beta(0.7, 0.3)	10.8	5.8	2.4	1.4	0.4	-
U3 - Beta(1, 1)	10.8	5.8	2.4	1.3	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	2.9	0.9	0.5	0.1	-
U2 - Beta(0.5, 0.5)	2.8	2.9	0.8	0.6	0.2	-
U2 - Beta(0.7, 0.3)	2.8	3	0.8	0.6	0.2	-
U2 - Beta(1, 1)	2.8	3	0.8	0.6	0.2	-
U3 - Beta(0.3, 0.7)	3	2.8	0.8	0.5	0.1	-
U3 - Beta(0.5, 0.5)	2.9	2.9	0.8	0.6	0.2	-
U3 - Beta(0.7, 0.3)	2.9	2.9	0.8	0.6	0.2	-
U3 - Beta(1, 1)	2.9	3	0.7	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						720.7
U2 - Beta(0.5, 0.5)						730.9
U2 - Beta(0.7, 0.3)						736.9
U2 - Beta(1, 1)						734.4
U3 - Beta(0.3, 0.7)						721.1
U3 - Beta(0.5, 0.5)						733.4
U3 - Beta(0.7, 0.3)						732.8
U3 - Beta(1, 1)						730
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.4	25.8	43.2	5.6	0.5	2.5
U2 - Beta(0.5, 0.5)	24.4	17.7	38.8	14.6	2.1	2.4
U2 - Beta(0.7, 0.3)	24.9	18.6	36.4	14.8	2.7	2.6
U2 - Beta(1, 1)	21.1	18.4	40.8	14.7	2.5	2.5
U3 - Beta(0.3, 0.7)	31.8	29.2	32.3	3.8	0.2	2.7
U3 - Beta(0.5, 0.5)	28.7	18.4	35.7	12.8	1.7	2.7
U3 - Beta(0.7, 0.3)	28.3	19.3	33.6	13.9	2.3	2.6
U3 - Beta(1, 1)	28.7	18.4	36.1	12.7	1.6	2.5
Mean administrations						
U2 - Beta(0.3, 0.7)	11.2	9.1	8	1.1	0.1	-
U2 - Beta(0.5, 0.5)	9	6.9	8.9	3.9	0.9	-
U2 - Beta(0.7, 0.3)	9	6.7	7.6	4.7	1.6	-
U2 - Beta(1, 1)	8.7	7	9.1	4	0.9	-
U3 - Beta(0.3, 0.7)	12.7	9.6	6.5	0.8	0	-
U3 - Beta(0.5, 0.5)	9.4	7.1	8.7	3.6	0.8	-
U3 - Beta(0.7, 0.3)	9.2	6.9	7.9	4.5	1.1	-
U3 - Beta(1, 1)	9.4	7.2	8.7	3.6	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	3.8	2.4	0.3	0	-
U2 - Beta(0.5, 0.5)	2.2	3.2	2.4	1.1	0.3	-
U2 - Beta(0.7, 0.3)	2.2	3.1	2	1.4	0.4	-
U2 - Beta(1, 1)	2.1	3.2	2.5	1.2	0.2	-
U3 - Beta(0.3, 0.7)	3.8	4	1.9	0.3	0	-
U3 - Beta(0.5, 0.5)	2.4	3.3	2.4	1.1	0.2	-
U3 - Beta(0.7, 0.3)	2.3	3.2	2.1	1.3	0.3	-
U3 - Beta(1, 1)	2.4	3.3	2.3	1.1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1090.1
U2 - Beta(0.5, 0.5)						1125.9
U2 - Beta(0.7, 0.3)						1130.3
U2 - Beta(1, 1)						1127.7
U3 - Beta(0.3, 0.7)						1079.9
U3 - Beta(0.5, 0.5)						1121.2
U3 - Beta(0.7, 0.3)						1125.8
U3 - Beta(1, 1)						1121

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.8	60.7	19.8	6.4	0.6	7.7
U2 - Beta(0.5, 0.5)	8.4	44.2	25.9	11.9	2.4	7.2
U2 - Beta(0.7, 0.3)	8.2	46.6	22.9	12.3	2.5	7.5
U2 - Beta(1, 1)	8.4	44	24.9	12.7	2.8	7.2
U3 - Beta(0.3, 0.7)	6.6	65.2	15.3	4.9	0	8
U3 - Beta(0.5, 0.5)	9.9	50	21.5	9.7	1.7	7.2
U3 - Beta(0.7, 0.3)	11	50.2	19.2	10.5	1.8	7.3
U3 - Beta(1, 1)	10.8	52.2	19	8.8	1.9	7.3
Mean administrations						
U2 - Beta(0.3, 0.7)	6.9	16.1	4.4	1.2	0.1	-
U2 - Beta(0.5, 0.5)	6.1	10.9	6.8	3.9	1.2	-
U2 - Beta(0.7, 0.3)	6.1	9.8	6.9	4.5	1.7	-
U2 - Beta(1, 1)	6	11	6.8	3.9	1.2	-
U3 - Beta(0.3, 0.7)	7.4	16.8	3.6	1	0.1	-
U3 - Beta(0.5, 0.5)	6.3	11.8	6.5	3.4	1	-
U3 - Beta(0.7, 0.3)	6.2	10.6	6.6	4.2	1.3	-
U3 - Beta(1, 1)	6.3	11.9	6.4	3.4	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.5	6.1	1.4	0.4	0.1	-
U2 - Beta(0.5, 0.5)	1.1	4.5	2	1.3	0.4	-
U2 - Beta(0.7, 0.3)	1.1	4.2	1.9	1.4	0.5	-
U2 - Beta(1, 1)	1	4.6	2	1.3	0.3	-
U3 - Beta(0.3, 0.7)	1.7	6.3	1.2	0.3	0	-
U3 - Beta(0.5, 0.5)	1.1	4.8	1.9	1.1	0.3	-
U3 - Beta(0.7, 0.3)	1.1	4.5	1.9	1.3	0.4	-
U3 - Beta(1, 1)	1.1	4.9	1.9	1.1	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1066.9
U2 - Beta(0.5, 0.5)						1085.5
U2 - Beta(0.7, 0.3)						1090.1
U2 - Beta(1, 1)						1085.7
U3 - Beta(0.3, 0.7)						1060
U3 - Beta(0.5, 0.5)						1081.9
U3 - Beta(0.7, 0.3)						1085.7
U3 - Beta(1, 1)						1079.8

Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.8	5.9	26.2	56.2	6.6	1.3
U2 - Beta(0.5, 0.5)	4.8	5.9	16.8	53	18.2	1.3
U2 - Beta(0.7, 0.3)	7.1	5.9	18	48	19.9	1.1
U2 - Beta(1, 1)	7.3	5.1	17.1	52	17.4	1.1
U3 - Beta(0.3, 0.7)	6.3	5.9	28.2	54	4.2	1.4
U3 - Beta(0.5, 0.5)	9.7	7.6	16.2	51.1	14.3	1.1
U3 - Beta(0.7, 0.3)	10.3	8	18.2	46.2	16.1	1.2
U3 - Beta(1, 1)	10.2	7.3	15.6	52.7	13	1.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.4	5.3	9.5	9.5	1	-
U2 - Beta(0.5, 0.5)	4.1	4.5	6.7	10.5	4	-
U2 - Beta(0.7, 0.3)	4.1	4.5	6.6	9.6	5.1	-
U2 - Beta(1, 1)	4.1	4.5	6.7	10.5	4	-
U3 - Beta(0.3, 0.7)	4.7	5.4	9.7	9.3	0.8	-
U3 - Beta(0.5, 0.5)	4.2	4.6	6.8	10.7	3.5	-
U3 - Beta(0.7, 0.3)	4.3	4.7	6.6	9.7	4.5	-
U3 - Beta(1, 1)	4.5	4.8	6.5	10.7	3.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.1	3.3	3	0.3	-
U2 - Beta(0.5, 0.5)	0.5	3	1.9	3.3	1.1	-
U2 - Beta(0.7, 0.3)	0.5	3	1.9	3.1	1.3	-
U2 - Beta(1, 1)	0.5	3	1.9	3.3	1.1	-
U3 - Beta(0.3, 0.7)	0.8	3.1	3.4	3	0.2	-
U3 - Beta(0.5, 0.5)	0.5	3	2	3.4	0.9	-
U3 - Beta(0.7, 0.3)	0.6	3.1	1.9	3.1	1.2	-
U3 - Beta(1, 1)	0.6	3.1	1.9	3.4	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1068
U2 - Beta(0.5, 0.5)						1107.9
U2 - Beta(0.7, 0.3)						1111.9
U2 - Beta(1, 1)						1107.3
U3 - Beta(0.3, 0.7)						1063.9
U3 - Beta(0.5, 0.5)						1104.9
U3 - Beta(0.7, 0.3)						1106
U3 - Beta(1, 1)						1101.1

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	1.4	39.7	27.6	21.3	8.5	1.5
U2 - Beta(0.5, 0.5)	5.8	35.9	26.1	19.9	10.9	1.4
U2 - Beta(0.7, 0.3)	7	37.4	25.9	20	8.3	1.4
U2 - Beta(1, 1)	7.3	38	25.7	18.2	9.4	1.4
U3 - Beta(0.3, 0.7)	3.5	48.5	23.8	17.4	5.2	1.6
U3 - Beta(0.5, 0.5)	8.4	43.5	23.6	17.3	5.8	1.4
U3 - Beta(0.7, 0.3)	8.2	42.5	25.3	15.4	7.2	1.4
U3 - Beta(1, 1)	11.3	45.2	21	15.3	5.8	1.4
Mean administrations						
U2 - Beta(0.3, 0.7)	4.2	12.9	7	4.2	1.5	-
U2 - Beta(0.5, 0.5)	3.8	7.1	7	7.2	4.7	-
U2 - Beta(0.7, 0.3)	3.8	6.5	7.1	7.5	4.8	-
U2 - Beta(1, 1)	3.8	7.2	7.2	7.1	4.5	-
U3 - Beta(0.3, 0.7)	4.4	14	6.6	3.6	1.2	-
U3 - Beta(0.5, 0.5)	3.9	7.8	7.1	6.9	4.1	-
U3 - Beta(0.7, 0.3)	4	7.3	7	7.1	4.4	-
U3 - Beta(1, 1)	3.9	8.1	7.1	6.9	3.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	5.7	2.6	1.5	0.6	-
U2 - Beta(0.5, 0.5)	0.3	3.8	2.2	2.7	1.7	-
U2 - Beta(0.7, 0.3)	0.3	3.7	2.2	2.9	1.6	-
U2 - Beta(1, 1)	0.3	3.8	2.3	2.7	1.6	-
U3 - Beta(0.3, 0.7)	0.6	6.1	2.5	1.4	0.4	-
U3 - Beta(0.5, 0.5)	0.3	4.1	2.3	2.6	1.4	-
U3 - Beta(0.7, 0.3)	0.3	3.9	2.3	2.7	1.5	-
U3 - Beta(1, 1)	0.3	4.1	2.3	2.6	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1035.9
U2 - Beta(0.5, 0.5)						1052.1
U2 - Beta(0.7, 0.3)						1054.4
U2 - Beta(1, 1)						1051.3
U3 - Beta(0.3, 0.7)						1033.2
U3 - Beta(0.5, 0.5)						1050.1
U3 - Beta(0.7, 0.3)						1052.7
U3 - Beta(1, 1)						1049
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	63.6	15	5.5	0.9	0.1	14.9
U2 - Beta(0.5, 0.5)	61	18.6	5.1	1.8	0.5	13
U2 - Beta(0.7, 0.3)	60.8	18.9	4.8	2	0.5	13
U2 - Beta(1, 1)	61.3	18.6	4.9	1.9	0.5	12.8
U3 - Beta(0.3, 0.7)	68.1	13.6	3.1	0.5	0.1	14.6
U3 - Beta(0.5, 0.5)	63.7	17.6	3.7	1.4	0.3	13.3
U3 - Beta(0.7, 0.3)	62.9	17.3	4.8	1	0.4	13.6
U3 - Beta(1, 1)	63.2	18.5	3.2	1.3	0.4	13.4
Mean administrations						
U2 - Beta(0.3, 0.7)	20.6	4.9	1.5	0.3	0	-
U2 - Beta(0.5, 0.5)	17.1	6.3	2.7	1.3	0.3	-
U2 - Beta(0.7, 0.3)	16.8	6.4	2.7	1.5	0.5	-
U2 - Beta(1, 1)	17.1	6.3	2.7	1.4	0.3	-
U3 - Beta(0.3, 0.7)	21.4	4.7	1	0.2	0	-
U3 - Beta(0.5, 0.5)	17.7	6.2	2.4	1.2	0.3	-
U3 - Beta(0.7, 0.3)	16.9	6.3	2.7	1.4	0.4	-
U3 - Beta(1, 1)	17.6	6.3	2.3	1.2	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.6	2	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	4.3	2.5	0.8	0.4	0.1	-
U2 - Beta(0.7, 0.3)	4.2	2.5	0.7	0.5	0.1	-
U2 - Beta(1, 1)	4.3	2.5	0.8	0.4	0.1	-
U3 - Beta(0.3, 0.7)	5.9	1.9	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	4.5	2.5	0.7	0.4	0.1	-
U3 - Beta(0.7, 0.3)	4.2	2.6	0.8	0.5	0.1	-
U3 - Beta(1, 1)	4.5	2.5	0.7	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1059.8
U2 - Beta(0.5, 0.5)						1081.6
U2 - Beta(0.7, 0.3)						1085.6
U2 - Beta(1, 1)						1081.6
U3 - Beta(0.3, 0.7)						1059.8
U3 - Beta(0.5, 0.5)						1079.7
U3 - Beta(0.7, 0.3)						1079.1
U3 - Beta(1, 1)						1076.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.7	69.3	12.8	2.5	0.2	2.5
U2 - Beta(0.5, 0.5)	11.8	46.5	23.6	12	3.3	2.8
U2 - Beta(0.7, 0.3)	11	44.9	23.3	13.9	4	2.9
U2 - Beta(1, 1)	10.6	47	24.3	11.4	3.9	2.8
U3 - Beta(0.3, 0.7)	16	71.4	8.4	1.3	0.2	2.7
U3 - Beta(0.5, 0.5)	14.3	49.8	21.2	9.8	2.1	2.8
U3 - Beta(0.7, 0.3)	14.1	45.3	22.8	12.5	2.5	2.8
U3 - Beta(1, 1)	14.2	52.9	19.1	9.2	1.9	2.7
Mean administrations						
U2 - Beta(0.3, 0.7)	10.4	16.2	2.5	0.4	0.1	-
U2 - Beta(0.5, 0.5)	7.4	12.5	6	2.9	0.8	-
U2 - Beta(0.7, 0.3)	7.1	9.5	6.4	4.6	1.9	-
U2 - Beta(1, 1)	7.2	12.5	6	3	0.9	-
U3 - Beta(0.3, 0.7)	11	16.4	1.9	0.3	0	-
U3 - Beta(0.5, 0.5)	7.7	13.1	5.5	2.5	0.7	-
U3 - Beta(0.7, 0.3)	7.4	10.5	6.4	4.1	1.3	-
U3 - Beta(1, 1)	7.7	13.3	5.3	2.5	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	5.5	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	1.4	4.6	1.6	0.9	0.2	-
U2 - Beta(0.7, 0.3)	1.3	3.8	1.6	1.3	0.6	-
U2 - Beta(1, 1)	1.4	4.6	1.6	0.9	0.2	-
U3 - Beta(0.3, 0.7)	3	5.5	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	1.6	4.8	1.4	0.8	0.2	-
U3 - Beta(0.7, 0.3)	1.4	4	1.7	1.1	0.4	-
U3 - Beta(1, 1)	1.6	4.9	1.4	0.7	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1129
U2 - Beta(0.5, 0.5)						1151.5
U2 - Beta(0.7, 0.3)						1157.3
U2 - Beta(1, 1)						1153
U3 - Beta(0.3, 0.7)						1124.7
U3 - Beta(0.5, 0.5)						1149.8
U3 - Beta(0.7, 0.3)						1154.3
U3 - Beta(1, 1)						1148.6
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.6	11.2	4.3	1.8	0.4	55.7
U2 - Beta(0.5, 0.5)	28.4	13.5	3.6	1.2	0.3	53
U2 - Beta(0.7, 0.3)	27.4	14.1	3.3	1.1	0.4	53.7
U2 - Beta(1, 1)	27.4	14.3	3.4	1.1	0.3	53.5
U3 - Beta(0.3, 0.7)	26.2	12.2	3.4	1.5	0.2	56.5
U3 - Beta(0.5, 0.5)	28	14.2	2.4	1.1	0.2	54.1
U3 - Beta(0.7, 0.3)	28.4	13.7	2.8	0.9	0.2	54
U3 - Beta(1, 1)	28.7	13.4	2.4	0.9	0.2	54.4
Mean administrations						
U2 - Beta(0.3, 0.7)	13.1	5.8	2.2	0.8	0.1	-
U2 - Beta(0.5, 0.5)	12.7	5.9	2.3	1.2	0.3	-
U2 - Beta(0.7, 0.3)	12.6	6	2.2	1.2	0.3	-
U2 - Beta(1, 1)	12.6	6.1	2.3	1.2	0.3	-
U3 - Beta(0.3, 0.7)	13.4	5.8	2	0.7	0.1	-
U3 - Beta(0.5, 0.5)	12.8	5.9	2.2	1.1	0.3	-
U3 - Beta(0.7, 0.3)	12.9	5.9	2.2	1.2	0.3	-
U3 - Beta(1, 1)	12.9	6	2.1	1.2	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	2.7	0.8	0.3	0	-
U2 - Beta(0.5, 0.5)	3.4	2.8	0.7	0.4	0.1	-
U2 - Beta(0.7, 0.3)	3.3	2.8	0.7	0.5	0.1	-
U2 - Beta(1, 1)	3.3	2.9	0.7	0.4	0.1	-
U3 - Beta(0.3, 0.7)	3.7	2.7	0.7	0.3	0	-
U3 - Beta(0.5, 0.5)	3.4	2.8	0.7	0.4	0.1	-
U3 - Beta(0.7, 0.3)	3.4	2.8	0.7	0.5	0.1	-
U3 - Beta(1, 1)	3.4	2.8	0.6	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						802
U2 - Beta(0.5, 0.5)						821.9
U2 - Beta(0.7, 0.3)						821.3
U2 - Beta(1, 1)						822.9
U3 - Beta(0.3, 0.7)						796.9
U3 - Beta(0.5, 0.5)						816.6
U3 - Beta(0.7, 0.3)						818.2
U3 - Beta(1, 1)						819.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.7	8	24.3	24.2	21.1	14.7
U2 - Beta(0.5, 0.5)	11.4	10.9	20.8	16	27.5	13.4
U2 - Beta(0.7, 0.3)	11.5	10.9	21.7	15.9	27.2	12.8
U2 - Beta(1, 1)	13.2	10.8	20.5	15.3	27	13.2
U3 - Beta(0.3, 0.7)	11.8	12.5	24	20.5	16.6	14.6
U3 - Beta(0.5, 0.5)	15.2	13.1	20.5	14.6	23.7	12.9
U3 - Beta(0.7, 0.3)	17.5	13.1	20.2	14	21.8	13.4
U3 - Beta(1, 1)	18.7	12.9	18.8	14.1	22	13.5
Mean administrations						
U2 - Beta(0.3, 0.7)	6.5	5.9	7.2	5.7	2.9	-
U2 - Beta(0.5, 0.5)	6.2	5.6	6	5.4	5.2	-
U2 - Beta(0.7, 0.3)	6.1	5.7	5.8	5.2	5.5	-
U2 - Beta(1, 1)	6.1	5.8	6	5.3	5.2	-
U3 - Beta(0.3, 0.7)	6.8	6.1	7.2	5.4	2.5	-
U3 - Beta(0.5, 0.5)	6.4	5.7	6	5.4	4.9	-
U3 - Beta(0.7, 0.3)	6.4	5.9	5.7	5.2	5.1	-
U3 - Beta(1, 1)	6.6	6	5.7	5.3	4.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.4	3.3	2.6	2.2	1	-
U2 - Beta(0.5, 0.5)	1.3	3.3	2	2.1	1.6	-
U2 - Beta(0.7, 0.3)	1.2	3.4	1.9	2	1.6	-
U2 - Beta(1, 1)	1.2	3.4	2	2.1	1.6	-
U3 - Beta(0.3, 0.7)	1.6	3.3	2.6	2.1	0.8	-
U3 - Beta(0.5, 0.5)	1.3	3.3	2	2.1	1.5	-
U3 - Beta(0.7, 0.3)	1.3	3.4	1.9	2.1	1.5	-
U3 - Beta(1, 1)	1.4	3.5	1.9	2.1	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						975.3
U2 - Beta(0.5, 0.5)						999.4
U2 - Beta(0.7, 0.3)						1002.3
U2 - Beta(1, 1)						999.6
U3 - Beta(0.3, 0.7)						970.5
U3 - Beta(0.5, 0.5)						997.6
U3 - Beta(0.7, 0.3)						997.5
U3 - Beta(1, 1)						994.2

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5	44.8	11.8	2.9	0.3	35.2
U2 - Beta(0.5, 0.5)	4.6	35.2	17.1	6.6	1.7	34.8
U2 - Beta(0.7, 0.3)	5	39.3	14.4	5.9	2	33.4
U2 - Beta(1, 1)	4.5	36.4	15.9	7.1	1.8	34.3
U3 - Beta(0.3, 0.7)	5.5	46.9	9.7	2.5	0.1	35.3
U3 - Beta(0.5, 0.5)	5.4	38.7	14.9	5.2	1	34.8
U3 - Beta(0.7, 0.3)	6.2	39.2	13.2	6	1.3	34.1
U3 - Beta(1, 1)	5.1	40.8	13.5	5	1.1	34.5
Mean administrations						
U2 - Beta(0.3, 0.7)	8	13.2	2.8	0.6	0.1	-
U2 - Beta(0.5, 0.5)	7.5	9.7	4.8	2.3	0.6	-
U2 - Beta(0.7, 0.3)	7.4	9	4.9	2.9	1	-
U2 - Beta(1, 1)	7.4	9.8	4.8	2.3	0.7	-
U3 - Beta(0.3, 0.7)	8.2	13.5	2.5	0.5	0.1	-
U3 - Beta(0.5, 0.5)	7.6	10.2	4.6	2	0.5	-
U3 - Beta(0.7, 0.3)	7.5	9.3	4.8	2.6	0.8	-
U3 - Beta(1, 1)	7.6	10.5	4.4	2	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	5.2	0.9	0.2	0	-
U2 - Beta(0.5, 0.5)	1.5	4.1	1.3	0.7	0.2	-
U2 - Beta(0.7, 0.3)	1.4	3.9	1.3	0.9	0.3	-
U2 - Beta(1, 1)	1.4	4.2	1.3	0.7	0.2	-
U3 - Beta(0.3, 0.7)	1.8	5.3	0.8	0.2	0	-
U3 - Beta(0.5, 0.5)	1.5	4.3	1.2	0.7	0.1	-
U3 - Beta(0.7, 0.3)	1.5	4	1.3	0.8	0.2	-
U3 - Beta(1, 1)	1.5	4.4	1.2	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						924.5
U2 - Beta(0.5, 0.5)						943.9
U2 - Beta(0.7, 0.3)						958
U2 - Beta(1, 1)						946.9
U3 - Beta(0.3, 0.7)						922.7
U3 - Beta(0.5, 0.5)						944.9
U3 - Beta(0.7, 0.3)						950.6
U3 - Beta(1, 1)						946.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	52	25.7	7.7	0.4	0	14.2
U2 - Beta(0.5, 0.5)	56.8	23.2	5.2	0.7	0	14.1
U2 - Beta(0.7, 0.3)	56.4	23.8	5.8	0.5	0.1	13.4
U2 - Beta(1, 1)	55.2	25.4	5.2	0.8	0.1	13.3
U3 - Beta(0.3, 0.7)	56.9	24.1	3.7	0.4	0	14.9
U3 - Beta(0.5, 0.5)	60.6	21.2	3.9	0.6	0	13.7
U3 - Beta(0.7, 0.3)	60.1	20.7	4.3	0.7	0.1	14.1
U3 - Beta(1, 1)	59.7	21.4	4.4	0.6	0.1	13.8
Mean administrations						
U2 - Beta(0.3, 0.7)	16.2	8.1	3	0.8	0.1	-
U2 - Beta(0.5, 0.5)	15.4	7.8	3.4	1.4	0.2	-
U2 - Beta(0.7, 0.3)	15.4	7.9	3.3	1.4	0.2	-
U2 - Beta(1, 1)	15.1	8	3.5	1.5	0.2	-
U3 - Beta(0.3, 0.7)	17.2	7.8	2.3	0.6	0.1	-
U3 - Beta(0.5, 0.5)	15.9	7.6	3.2	1.3	0.2	-
U3 - Beta(0.7, 0.3)	15.7	7.8	3.2	1.4	0.2	-
U3 - Beta(1, 1)	16.1	7.8	3	1.3	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.8	3.4	1	0.3	0	-
U2 - Beta(0.5, 0.5)	4.4	3.5	1	0.5	0.1	-
U2 - Beta(0.7, 0.3)	4.4	3.5	1	0.5	0.1	-
U2 - Beta(1, 1)	4.3	3.6	1	0.5	0.1	-
U3 - Beta(0.3, 0.7)	5.2	3.4	0.8	0.2	0	-
U3 - Beta(0.5, 0.5)	4.6	3.5	0.9	0.5	0.1	-
U3 - Beta(0.7, 0.3)	4.5	3.5	0.9	0.5	0.1	-
U3 - Beta(1, 1)	4.6	3.6	0.8	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1020.7
U2 - Beta(0.5, 0.5)						1033.8
U2 - Beta(0.7, 0.3)						1035.8
U2 - Beta(1, 1)						1035.5
U3 - Beta(0.3, 0.7)						1015.6
U3 - Beta(0.5, 0.5)						1033.3
U3 - Beta(0.7, 0.3)						1033.9
U3 - Beta(1, 1)						1032.5
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.1	10.1	9.5	58.7	12.7	1.9
U2 - Beta(0.5, 0.5)	10.4	11.4	8.2	44.3	23.5	2.2
U2 - Beta(0.7, 0.3)	11.4	12.6	8.1	41.1	24.8	2
U2 - Beta(1, 1)	11.9	10.5	7.3	42.5	25.9	1.9
U3 - Beta(0.3, 0.7)	11.1	11.2	6.9	58.3	10.2	2.3
U3 - Beta(0.5, 0.5)	14.8	13.7	6.9	41.1	21.5	2
U3 - Beta(0.7, 0.3)	17.2	14	6	39	21.7	2.1
U3 - Beta(1, 1)	18.2	13.5	6.3	39.2	20.7	2.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	5.6	6	11.3	2.3	-
U2 - Beta(0.5, 0.5)	4.4	5	5.5	9.4	5.5	-
U2 - Beta(0.7, 0.3)	4.5	5	5.4	8.6	6.3	-
U2 - Beta(1, 1)	4.4	5	5.4	9.4	5.6	-
U3 - Beta(0.3, 0.7)	4.8	5.9	6	11.1	2	-
U3 - Beta(0.5, 0.5)	4.6	5.1	5.4	9.5	5.2	-
U3 - Beta(0.7, 0.3)	4.7	5.2	5.3	8.8	5.7	-
U3 - Beta(1, 1)	4.8	5.2	5.2	9.4	5.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.3	2.1	3.9	0.7	-
U2 - Beta(0.5, 0.5)	0.6	3.2	1.6	3.3	1.4	-
U2 - Beta(0.7, 0.3)	0.6	3.2	1.6	3.1	1.5	-
U2 - Beta(1, 1)	0.6	3.2	1.6	3.3	1.4	-
U3 - Beta(0.3, 0.7)	0.9	3.4	2.1	3.8	0.5	-
U3 - Beta(0.5, 0.5)	0.7	3.2	1.6	3.3	1.3	-
U3 - Beta(0.7, 0.3)	0.7	3.3	1.6	3.1	1.4	-
U3 - Beta(1, 1)	0.8	3.3	1.6	3.2	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1055.1
U2 - Beta(0.5, 0.5)						1084.3
U2 - Beta(0.7, 0.3)						1090.1
U2 - Beta(1, 1)						1084.4
U3 - Beta(0.3, 0.7)						1048.1
U3 - Beta(0.5, 0.5)						1081
U3 - Beta(0.7, 0.3)						1083.4
U3 - Beta(1, 1)						1079.1

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.1	36.6	16.4	6.4	1.1	33.4
U2 - Beta(0.5, 0.5)	6	37	17.9	6.4	1.8	30.9
U2 - Beta(0.7, 0.3)	6.1	36.8	17	7.3	2.1	30.7
U2 - Beta(1, 1)	7.5	36.5	16.7	6.6	1.6	31.1
U3 - Beta(0.3, 0.7)	6.8	40.3	13.4	4.6	0.7	34.2
U3 - Beta(0.5, 0.5)	7.8	38.4	15.1	5.4	1.2	32.1
U3 - Beta(0.7, 0.3)	9.6	37.3	14.9	5.6	1.1	31.5
U3 - Beta(1, 1)	9.7	37.8	14.8	4.9	1.1	31.7
Mean administrations						
U2 - Beta(0.3, 0.7)	8	11.4	4	1.5	0.3	-
U2 - Beta(0.5, 0.5)	7.6	9.1	5.1	2.9	0.9	-
U2 - Beta(0.7, 0.3)	7.5	8.8	4.9	3.1	1.2	-
U2 - Beta(1, 1)	7.5	9.1	5	3	0.9	-
U3 - Beta(0.3, 0.7)	8.2	12	3.5	1.2	0.2	-
U3 - Beta(0.5, 0.5)	7.7	9.3	4.9	2.7	0.8	-
U3 - Beta(0.7, 0.3)	7.6	9.2	4.8	3.1	0.9	-
U3 - Beta(1, 1)	7.7	9.6	4.8	2.6	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	4.9	1.3	0.5	0.1	-
U2 - Beta(0.5, 0.5)	1.5	4.2	1.5	1	0.3	-
U2 - Beta(0.7, 0.3)	1.5	4.1	1.4	1.1	0.3	-
U2 - Beta(1, 1)	1.5	4.2	1.5	1	0.3	-
U3 - Beta(0.3, 0.7)	1.8	5.1	1.2	0.4	0.1	-
U3 - Beta(0.5, 0.5)	1.6	4.3	1.4	1	0.3	-
U3 - Beta(0.7, 0.3)	1.5	4.2	1.4	1	0.3	-
U3 - Beta(1, 1)	1.6	4.4	1.4	0.9	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						912
U2 - Beta(0.5, 0.5)						934.4
U2 - Beta(0.7, 0.3)						938.4
U2 - Beta(1, 1)						932.8
U3 - Beta(0.3, 0.7)						906.3
U3 - Beta(0.5, 0.5)						930.2
U3 - Beta(0.7, 0.3)						934.1
U3 - Beta(1, 1)						929.9
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.8	13.7	14.7	40	15	5.8
U2 - Beta(0.5, 0.5)	15.2	14.9	14.4	34.2	15.8	5.5
U2 - Beta(0.7, 0.3)	16.2	15.7	15	32	15.7	5.4
U2 - Beta(1, 1)	18	13.4	14.5	33.2	15.6	5.3
U3 - Beta(0.3, 0.7)	15.5	15.7	13.2	39.3	10.7	5.6
U3 - Beta(0.5, 0.5)	20.9	17.3	14.5	30.2	11.5	5.6
U3 - Beta(0.7, 0.3)	22.4	17.4	13.7	29.4	11.7	5.4
U3 - Beta(1, 1)	24	17.1	13.7	29.1	10.7	5.4
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	6.3	6.2	8.5	2.7	-
U2 - Beta(0.5, 0.5)	5.7	5.9	5.8	7.7	4.3	-
U2 - Beta(0.7, 0.3)	5.7	6	5.7	7.4	4.6	-
U2 - Beta(1, 1)	5.7	6	5.7	7.7	4.3	-
U3 - Beta(0.3, 0.7)	6.1	6.5	6.1	8.5	2.3	-
U3 - Beta(0.5, 0.5)	6	6.1	5.6	7.7	4	-
U3 - Beta(0.7, 0.3)	6.1	6.1	5.6	7.5	4.2	-
U3 - Beta(1, 1)	6.2	6.3	5.5	7.5	4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.2	3.5	2.2	3.1	0.9	-
U2 - Beta(0.5, 0.5)	1.2	3.5	1.9	2.9	1.4	-
U2 - Beta(0.7, 0.3)	1.1	3.6	1.8	2.9	1.4	-
U2 - Beta(1, 1)	1.1	3.6	1.8	2.9	1.4	-
U3 - Beta(0.3, 0.7)	1.4	3.6	2.2	3.1	0.8	-
U3 - Beta(0.5, 0.5)	1.3	3.6	1.8	2.9	1.3	-
U3 - Beta(0.7, 0.3)	1.3	3.6	1.8	2.8	1.4	-
U3 - Beta(1, 1)	1.4	3.7	1.7	2.8	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1012.1
U2 - Beta(0.5, 0.5)						1027.4
U2 - Beta(0.7, 0.3)						1030.2
U2 - Beta(1, 1)						1028.2
U3 - Beta(0.3, 0.7)						1009.7
U3 - Beta(0.5, 0.5)						1024.5
U3 - Beta(0.7, 0.3)						1026.8
U3 - Beta(1, 1)						1023.3

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	37.1	29.7	14.4	6.4	0.8	11.6
U2 - Beta(0.5, 0.5)	41.1	28	12.2	6.7	0.4	11.6
U2 - Beta(0.7, 0.3)	40.6	26.5	13.6	6.6	1.1	11.6
U2 - Beta(1, 1)	41.2	25.5	13.6	7.6	1	11.1
U3 - Beta(0.3, 0.7)	44	31.9	8.7	3.1	0.3	12
U3 - Beta(0.5, 0.5)	46.3	27.7	10.4	3.7	0.3	11.6
U3 - Beta(0.7, 0.3)	47.4	26.6	9.9	3.7	0.3	12.1
U3 - Beta(1, 1)	45.3	27.2	11.4	3.9	0.3	11.9
Mean administrations						
U2 - Beta(0.3, 0.7)	9.3	8.2	6.1	4	1.3	-
U2 - Beta(0.5, 0.5)	9.4	8	5.8	4.1	1.5	-
U2 - Beta(0.7, 0.3)	9.4	8	5.8	4.1	1.6	-
U2 - Beta(1, 1)	9.3	8.1	5.8	4.1	1.5	-
U3 - Beta(0.3, 0.7)	10.3	8.3	5.5	3.6	1.1	-
U3 - Beta(0.5, 0.5)	9.9	8.1	5.5	3.9	1.5	-
U3 - Beta(0.7, 0.3)	10	8.1	5.5	3.8	1.4	-
U3 - Beta(1, 1)	10.1	8.2	5.4	3.8	1.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	4.1	2.2	1.6	0.5	-
U2 - Beta(0.5, 0.5)	2.6	4.2	1.9	1.7	0.5	-
U2 - Beta(0.7, 0.3)	2.7	4.1	1.9	1.7	0.5	-
U2 - Beta(1, 1)	2.6	4.2	1.9	1.7	0.5	-
U3 - Beta(0.3, 0.7)	3.1	4.1	1.9	1.5	0.4	-
U3 - Beta(0.5, 0.5)	2.9	4.2	1.8	1.7	0.5	-
U3 - Beta(0.7, 0.3)	2.9	4.2	1.8	1.6	0.5	-
U3 - Beta(1, 1)	2.9	4.2	1.8	1.6	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						977.2
U2 - Beta(0.5, 0.5)						982.1
U2 - Beta(0.7, 0.3)						982.3
U2 - Beta(1, 1)						983.4
U3 - Beta(0.3, 0.7)						976.9
U3 - Beta(0.5, 0.5)						980.9
U3 - Beta(0.7, 0.3)						982.4
U3 - Beta(1, 1)						981.3
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.9	18.7	21.2	22.1	17.1	0
U2 - Beta(0.5, 0.5)	41	24.4	14.9	13.3	6.4	0
U2 - Beta(0.7, 0.3)	41.1	24.5	16.2	11.5	6.7	0
U2 - Beta(1, 1)	43.3	23.3	15.4	11.3	6.7	0
U3 - Beta(0.3, 0.7)	29.4	24	20	17.3	9.3	0
U3 - Beta(0.5, 0.5)	45.3	24.8	13.4	11.9	4.6	0
U3 - Beta(0.7, 0.3)	46.3	25.7	14.4	9.1	4.5	0
U3 - Beta(1, 1)	50.4	24.4	13.6	7.7	3.9	0
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	6.6	6	6.1	3.6	-
U2 - Beta(0.5, 0.5)	4.4	4.9	6	8.3	6.5	-
U2 - Beta(0.7, 0.3)	4.4	4.8	6.1	8.3	6.4	-
U2 - Beta(1, 1)	4.5	5	6.3	8.2	6	-
U3 - Beta(0.3, 0.7)	8.4	7.1	6	5.6	2.9	-
U3 - Beta(0.5, 0.5)	4.7	5.1	6.2	8.3	5.7	-
U3 - Beta(0.7, 0.3)	5	5.4	6.4	7.9	5.4	-
U3 - Beta(1, 1)	5	5.4	6.4	8	5.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	3.3	2.5	2.5	1.5	-
U2 - Beta(0.5, 0.5)	0.6	2.8	2.2	3.6	2.4	-
U2 - Beta(0.7, 0.3)	0.6	2.9	2.2	3.7	2.3	-
U2 - Beta(1, 1)	0.6	2.9	2.3	3.6	2.2	-
U3 - Beta(0.3, 0.7)	2.4	3.5	2.5	2.4	1.2	-
U3 - Beta(0.5, 0.5)	0.7	3	2.3	3.6	2.1	-
U3 - Beta(0.7, 0.3)	0.8	3.1	2.3	3.5	1.9	-
U3 - Beta(1, 1)	0.8	3.1	2.4	3.5	1.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						994.8
U2 - Beta(0.5, 0.5)						1010.1
U2 - Beta(0.7, 0.3)						1012.3
U2 - Beta(1, 1)						1008.6
U3 - Beta(0.3, 0.7)						992.8
U3 - Beta(0.5, 0.5)						1007.6
U3 - Beta(0.7, 0.3)						1008
U3 - Beta(1, 1)						1005.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	34	57.5	7.2	0.6	0	0.7
U2 - Beta(0.5, 0.5)	19.4	59.5	17.7	2.7	0.1	0.6
U2 - Beta(0.7, 0.3)	21.8	54	19.8	3.8	0	0.6
U2 - Beta(1, 1)	18.6	60.1	18.4	2.4	0.1	0.4
U3 - Beta(0.3, 0.7)	39.6	55.1	4.3	0.4	0	0.6
U3 - Beta(0.5, 0.5)	23.7	60.5	13.7	1.6	0	0.5
U3 - Beta(0.7, 0.3)	24	56.1	16.3	3	0	0.6
U3 - Beta(1, 1)	22	61.1	14.4	1.9	0	0.6
Mean administrations						
U2 - Beta(0.3, 0.7)	15.5	12.5	1.7	0.1	0	-
U2 - Beta(0.5, 0.5)	8.9	14.2	5.3	1.3	0.2	-
U2 - Beta(0.7, 0.3)	8.9	11.9	6.3	2.5	0.4	-
U2 - Beta(1, 1)	8.7	14.2	5.4	1.3	0.2	-
U3 - Beta(0.3, 0.7)	16.4	12.3	1.1	0.1	0	-
U3 - Beta(0.5, 0.5)	9.6	14.5	4.7	1	0.1	-
U3 - Beta(0.7, 0.3)	9.2	12.7	5.8	2	0.2	-
U3 - Beta(1, 1)	9.5	14.6	4.6	1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.6	4.2	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	1.9	4.7	1.3	0.4	0	-
U2 - Beta(0.7, 0.3)	1.9	4.1	1.4	0.7	0.1	-
U2 - Beta(1, 1)	1.8	4.7	1.4	0.4	0.1	-
U3 - Beta(0.3, 0.7)	4.9	4.1	0.3	0	0	-
U3 - Beta(0.5, 0.5)	2.2	4.8	1.2	0.3	0	-
U3 - Beta(0.7, 0.3)	2	4.4	1.4	0.5	0.1	-
U3 - Beta(1, 1)	2.1	4.9	1.1	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1137.5
U2 - Beta(0.5, 0.5)						1189.2
U2 - Beta(0.7, 0.3)						1200
U2 - Beta(1, 1)						1191.2
U3 - Beta(0.3, 0.7)						1131.4
U3 - Beta(0.5, 0.5)						1183.3
U3 - Beta(0.7, 0.3)						1191.4
U3 - Beta(1, 1)						1182.1
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.3	7	9.9	22.5	56.3	0
U2 - Beta(0.5, 0.5)	10.9	9.7	9.3	14.9	55.2	0
U2 - Beta(0.7, 0.3)	12.4	10.9	11.2	13.2	52.3	0
U2 - Beta(1, 1)	14	10	8.1	12.9	55	0
U3 - Beta(0.3, 0.7)	9.1	9.9	11.5	21.5	48	0
U3 - Beta(0.5, 0.5)	17.9	13.9	9.4	11.5	47.3	0
U3 - Beta(0.7, 0.3)	19.5	14.6	11.1	10.4	44.4	0
U3 - Beta(1, 1)	21.6	14.7	9.1	9.5	45.1	0
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.7	5.9	7.7	7.8	-
U2 - Beta(0.5, 0.5)	3.5	4	5.2	6.7	10.6	-
U2 - Beta(0.7, 0.3)	3.5	4	5.1	6.5	10.8	-
U2 - Beta(1, 1)	3.5	4.2	5.1	6.6	10.6	-
U3 - Beta(0.3, 0.7)	3.9	4.9	6.2	7.8	7.1	-
U3 - Beta(0.5, 0.5)	3.6	4.2	5.2	6.8	10.2	-
U3 - Beta(0.7, 0.3)	3.8	4.4	5.2	6.5	10.1	-
U3 - Beta(1, 1)	3.8	4.5	5.2	6.6	9.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.9	2.3	3.1	2.4	-
U2 - Beta(0.5, 0.5)	0.2	2.8	1.8	2.8	3	-
U2 - Beta(0.7, 0.3)	0.3	2.8	1.8	2.8	3	-
U2 - Beta(1, 1)	0.3	2.9	1.8	2.8	3	-
U3 - Beta(0.3, 0.7)	0.5	3	2.5	3.1	2.2	-
U3 - Beta(0.5, 0.5)	0.3	2.9	1.8	2.8	2.9	-
U3 - Beta(0.7, 0.3)	0.4	3	1.8	2.7	2.8	-
U3 - Beta(1, 1)	0.4	3	1.8	2.8	2.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1035.7
U2 - Beta(0.5, 0.5)						1067
U2 - Beta(0.7, 0.3)						1068.7
U2 - Beta(1, 1)						1065.8
U3 - Beta(0.3, 0.7)						1029.2
U3 - Beta(0.5, 0.5)						1061.9
U3 - Beta(0.7, 0.3)						1060.3
U3 - Beta(1, 1)						1058.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.4	56.9	20.5	1.9	0	12.3
U2 - Beta(0.5, 0.5)	11.1	53.5	19.9	2.7	0.2	12.6
U2 - Beta(0.7, 0.3)	11.7	52.7	19.7	3.3	0	12.6
U2 - Beta(1, 1)	11.2	53.3	19.6	3.4	0	12.5
U3 - Beta(0.3, 0.7)	11.5	59.4	15.4	1.3	0	12.4
U3 - Beta(0.5, 0.5)	13.2	54.9	17	2.2	0.1	12.6
U3 - Beta(0.7, 0.3)	13.2	55	16.7	2.6	0	12.5
U3 - Beta(1, 1)	13.7	54.3	16.2	3.2	0.1	12.5
Mean administrations						
U2 - Beta(0.3, 0.7)	7.3	14.7	5.4	1.1	0.1	-
U2 - Beta(0.5, 0.5)	7.1	12.2	6.6	2.3	0.4	-
U2 - Beta(0.7, 0.3)	7.2	11.9	6.4	2.6	0.5	-
U2 - Beta(1, 1)	7.1	12.1	6.6	2.3	0.4	-
U3 - Beta(0.3, 0.7)	7.8	15.2	4.7	0.8	0.1	-
U3 - Beta(0.5, 0.5)	7.4	12.4	6.3	2.1	0.4	-
U3 - Beta(0.7, 0.3)	7.4	12.2	6.1	2.5	0.4	-
U3 - Beta(1, 1)	7.5	12.6	6.1	2.1	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.6	6	1.7	0.4	0	-
U2 - Beta(0.5, 0.5)	1.4	5.3	1.9	0.8	0.1	-
U2 - Beta(0.7, 0.3)	1.5	5.2	1.8	0.9	0.1	-
U2 - Beta(1, 1)	1.4	5.3	1.9	0.8	0.1	-
U3 - Beta(0.3, 0.7)	1.9	6.2	1.5	0.3	0	-
U3 - Beta(0.5, 0.5)	1.5	5.4	1.8	0.7	0.1	-
U3 - Beta(0.7, 0.3)	1.5	5.3	1.7	0.8	0.1	-
U3 - Beta(1, 1)	1.6	5.5	1.7	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1032.3
U2 - Beta(0.5, 0.5)						1048.1
U2 - Beta(0.7, 0.3)						1054.3
U2 - Beta(1, 1)						1047.4
U3 - Beta(0.3, 0.7)						1024.6
U3 - Beta(0.5, 0.5)						1043.6
U3 - Beta(0.7, 0.3)						1048.4
U3 - Beta(1, 1)						1044.3

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.5	44.9	30.9	14.5	1.8	2.4
U2 - Beta(0.5, 0.5)	11	46.5	26.6	11.7	2.2	2
U2 - Beta(0.7, 0.3)	10.1	44.5	27.8	14.3	1.4	1.9
U2 - Beta(1, 1)	11.4	45.1	25.8	13.5	2.1	2.1
U3 - Beta(0.3, 0.7)	8.7	50.2	27.4	10.1	1.1	2.5
U3 - Beta(0.5, 0.5)	13.1	48.1	24.1	11.6	1.2	1.9
U3 - Beta(0.7, 0.3)	13.3	49.9	21.7	11.6	1.4	2.1
U3 - Beta(1, 1)	15.9	49.5	19.4	11.8	1.3	2.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.7	12.5	7.8	3.8	0.9	-
U2 - Beta(0.5, 0.5)	4.7	9.8	8.2	5.2	1.8	-
U2 - Beta(0.7, 0.3)	4.8	9.6	8	5.3	2	-
U2 - Beta(1, 1)	4.8	9.8	8.1	5.2	1.8	-
U3 - Beta(0.3, 0.7)	5.1	13.6	7.4	3	0.6	-
U3 - Beta(0.5, 0.5)	4.9	10.4	7.8	4.9	1.7	-
U3 - Beta(0.7, 0.3)	5	10.2	7.7	5	1.7	-
U3 - Beta(1, 1)	5.1	10.7	7.5	4.8	1.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	5.6	2.7	1.4	0.3	-
U2 - Beta(0.5, 0.5)	0.7	4.8	2.6	1.8	0.6	-
U2 - Beta(0.7, 0.3)	0.7	4.7	2.4	1.9	0.6	-
U2 - Beta(1, 1)	0.7	4.8	2.6	1.8	0.6	-
U3 - Beta(0.3, 0.7)	0.9	6.1	2.6	1.1	0.2	-
U3 - Beta(0.5, 0.5)	0.8	5	2.5	1.7	0.6	-
U3 - Beta(0.7, 0.3)	0.8	5	2.4	1.8	0.6	-
U3 - Beta(1, 1)	0.9	5.1	2.4	1.7	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1039.6
U2 - Beta(0.5, 0.5)						1059.1
U2 - Beta(0.7, 0.3)						1059.8
U2 - Beta(1, 1)						1059
U3 - Beta(0.3, 0.7)						1033.9
U3 - Beta(0.5, 0.5)						1055.6
U3 - Beta(0.7, 0.3)						1058.5
U3 - Beta(1, 1)						1055.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	64.7	0.4	0	0	0	34.9
U2 - Beta(0.5, 0.5)	58.7	2	0	0	0	39.3
U2 - Beta(0.7, 0.3)	58.2	1.8	0	0	0	40
U2 - Beta(1, 1)	58.1	2.2	0	0	0	39.7
U3 - Beta(0.3, 0.7)	64.5	0.6	0	0	0	34.9
U3 - Beta(0.5, 0.5)	60	1.4	0	0	0	38.6
U3 - Beta(0.7, 0.3)	58.4	2	0	0	0	39.6
U3 - Beta(1, 1)	59.2	1.6	0	0	0	39.2
Mean administrations						
U2 - Beta(0.3, 0.7)	23.2	0.7	0	0	0	-
U2 - Beta(0.5, 0.5)	20.3	2.6	0.3	0	0	-
U2 - Beta(0.7, 0.3)	19.1	3.6	0.4	0.1	0	-
U2 - Beta(1, 1)	20.2	2.7	0.3	0	0	-
U3 - Beta(0.3, 0.7)	23.2	0.7	0	0	0	-
U3 - Beta(0.5, 0.5)	20.6	2.5	0.2	0	0	-
U3 - Beta(0.7, 0.3)	19.9	3.1	0.3	0	0	-
U3 - Beta(1, 1)	20.5	2.5	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.9	0.3	0	0	0	-
U2 - Beta(0.5, 0.5)	3.8	1	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.5	1.2	0.1	0	0	-
U2 - Beta(1, 1)	3.8	1	0.1	0	0	-
U3 - Beta(0.3, 0.7)	4.9	0.3	0	0	0	-
U3 - Beta(0.5, 0.5)	3.9	1	0	0	0	-
U3 - Beta(0.7, 0.3)	3.7	1.2	0.1	0	0	-
U3 - Beta(1, 1)	3.9	1	0	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1035
U2 - Beta(0.5, 0.5)						1011
U2 - Beta(0.7, 0.3)						1008.2
U2 - Beta(1, 1)						1008.2
U3 - Beta(0.3, 0.7)						1034.8
U3 - Beta(0.5, 0.5)						1013.8
U3 - Beta(0.7, 0.3)						1016.3
U3 - Beta(1, 1)						1012.1
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.7	59	14.8	3.1	0.2	10.2
U2 - Beta(0.5, 0.5)	13.5	57.6	13.8	3.9	0	11.2
U2 - Beta(0.7, 0.3)	13.6	57.7	13	4.4	0	11.3
U2 - Beta(1, 1)	15.4	56.3	13.7	3.7	0.1	10.8
U3 - Beta(0.3, 0.7)	15.1	60.5	11.1	2.5	0.2	10.6
U3 - Beta(0.5, 0.5)	16.1	57.8	11.2	3.3	0	11.6
U3 - Beta(0.7, 0.3)	16.7	55.2	12.3	4.6	0	11.2
U3 - Beta(1, 1)	18.2	54.3	12.1	4.5	0	10.9
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	14.6	5.3	1.4	0.1	-
U2 - Beta(0.5, 0.5)	7.9	12.9	5.8	2.2	0.2	-
U2 - Beta(0.7, 0.3)	7.9	12.5	5.8	2.4	0.3	-
U2 - Beta(1, 1)	7.8	12.8	5.9	2.3	0.2	-
U3 - Beta(0.3, 0.7)	8.2	15	4.7	1.1	0	-
U3 - Beta(0.5, 0.5)	8.2	12.9	5.6	2	0.2	-
U3 - Beta(0.7, 0.3)	8.2	12.7	5.6	2.3	0.2	-
U3 - Beta(1, 1)	8.3	12.9	5.5	2.1	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	6.1	1.8	0.4	0	-
U2 - Beta(0.5, 0.5)	1.8	5.7	1.6	0.7	0.1	-
U2 - Beta(0.7, 0.3)	1.8	5.6	1.6	0.8	0.1	-
U2 - Beta(1, 1)	1.8	5.6	1.7	0.7	0.1	-
U3 - Beta(0.3, 0.7)	2.1	6.2	1.5	0.4	0	-
U3 - Beta(0.5, 0.5)	2	5.7	1.6	0.7	0.1	-
U3 - Beta(0.7, 0.3)	1.9	5.6	1.5	0.8	0.1	-
U3 - Beta(1, 1)	2	5.7	1.6	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1037.3
U2 - Beta(0.5, 0.5)						1046.2
U2 - Beta(0.7, 0.3)						1048.8
U2 - Beta(1, 1)						1045.2
U3 - Beta(0.3, 0.7)						1030.2
U3 - Beta(0.5, 0.5)						1039.5
U3 - Beta(0.7, 0.3)						1043.5
U3 - Beta(1, 1)						1041.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.6	1.4	0	0	0	90
U2 - Beta(0.5, 0.5)	8.3	2.1	0.1	0	0	89.5
U2 - Beta(0.7, 0.3)	8.3	2	0	0	0	89.7
U2 - Beta(1, 1)	8.5	1.7	0.1	0	0	89.7
U3 - Beta(0.3, 0.7)	7.9	1.3	0	0	0	90.8
U3 - Beta(0.5, 0.5)	8.3	1.9	0.1	0	0	89.7
U3 - Beta(0.7, 0.3)	8.3	1.6	0.1	0	0	90
U3 - Beta(1, 1)	8.5	1.4	0.2	0	0	89.9
Mean administrations						
U2 - Beta(0.3, 0.7)	11.3	4.2	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	11.1	4.4	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	11.2	4.4	0.7	0.1	0	-
U2 - Beta(1, 1)	11.1	4.5	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	11.3	4.2	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	11.2	4.4	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	11.1	4.4	0.7	0.1	0	-
U3 - Beta(1, 1)	11.1	4.4	0.7	0.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	2.2	0.2	0	0	-
U2 - Beta(0.5, 0.5)	2.9	2.3	0.2	0	0	-
U2 - Beta(0.7, 0.3)	2.8	2.3	0.2	0	0	-
U2 - Beta(1, 1)	2.8	2.3	0.2	0	0	-
U3 - Beta(0.3, 0.7)	3	2.2	0.2	0	0	-
U3 - Beta(0.5, 0.5)	2.9	2.3	0.2	0	0	-
U3 - Beta(0.7, 0.3)	2.8	2.3	0.2	0	0	-
U3 - Beta(1, 1)	2.8	2.3	0.2	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						594.4
U2 - Beta(0.5, 0.5)						594.3
U2 - Beta(0.7, 0.3)						595.3
U2 - Beta(1, 1)						596.4
U3 - Beta(0.3, 0.7)						591.5
U3 - Beta(0.5, 0.5)						594.6
U3 - Beta(0.7, 0.3)						593.7
U3 - Beta(1, 1)						596.2
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.2	1	7.3	72.2	19.3	0
U2 - Beta(0.5, 0.5)	2.4	1.4	3.4	55.2	37.5	0.1
U2 - Beta(0.7, 0.3)	3.8	3.2	7.3	47.7	37.9	0.1
U2 - Beta(1, 1)	3	2.1	4.7	53.3	36.9	0
U3 - Beta(0.3, 0.7)	0.9	1.3	8.8	74.3	14.7	0
U3 - Beta(0.5, 0.5)	4	3.1	5.3	57.7	29.9	0
U3 - Beta(0.7, 0.3)	5.4	5.2	7	50.8	31.6	0
U3 - Beta(1, 1)	4.5	4.6	5.7	54.4	30.7	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	4	6.4	13.5	2.5	-
U2 - Beta(0.5, 0.5)	3.1	3.1	3.9	12.4	7.5	-
U2 - Beta(0.7, 0.3)	3.1	3.2	3.9	10.9	9.1	-
U2 - Beta(1, 1)	3.1	3.2	3.9	12.3	7.5	-
U3 - Beta(0.3, 0.7)	3.6	4	6.7	13.6	2.1	-
U3 - Beta(0.5, 0.5)	3.1	3.2	4	12.9	6.8	-
U3 - Beta(0.7, 0.3)	3.2	3.2	4.1	11.3	8.3	-
U3 - Beta(1, 1)	3.1	3.2	4.1	12.9	6.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.6	2.5	4.6	0.8	-
U2 - Beta(0.5, 0.5)	0.1	2.4	1.2	4.3	2.2	-
U2 - Beta(0.7, 0.3)	0	2.4	1.2	3.9	2.5	-
U2 - Beta(1, 1)	0	2.5	1.2	4.3	2.2	-
U3 - Beta(0.3, 0.7)	0.3	2.6	2.6	4.7	0.7	-
U3 - Beta(0.5, 0.5)	0	2.4	1.2	4.5	2	-
U3 - Beta(0.7, 0.3)	0.1	2.5	1.3	4.1	2.4	-
U3 - Beta(1, 1)	0	2.5	1.3	4.5	2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1051.3
U2 - Beta(0.5, 0.5)						1089.4
U2 - Beta(0.7, 0.3)						1094.9
U2 - Beta(1, 1)						1089.9
U3 - Beta(0.3, 0.7)						1049
U3 - Beta(0.5, 0.5)						1089.2
U3 - Beta(0.7, 0.3)						1089.5
U3 - Beta(1, 1)						1087.3

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.1	32.8	41.4	2.3	0.3	1.1
U2 - Beta(0.5, 0.5)	26.8	29	35.8	7.1	0.1	1.2
U2 - Beta(0.7, 0.3)	25.7	29.8	35.9	7	0.1	1.5
U2 - Beta(1, 1)	26.4	29.2	36.7	6.4	0.1	1.2
U3 - Beta(0.3, 0.7)	29.7	36.3	30.8	1.8	0.3	1.1
U3 - Beta(0.5, 0.5)	31.5	29.2	32	6	0	1.3
U3 - Beta(0.7, 0.3)	31.1	29	31.3	7.1	0.1	1.4
U3 - Beta(1, 1)	30.7	30.9	30.3	6.8	0	1.3
Mean administrations						
U2 - Beta(0.3, 0.7)	9.9	10.3	8.5	1.1	0.1	-
U2 - Beta(0.5, 0.5)	8.8	9.1	8.7	2.9	0.4	-
U2 - Beta(0.7, 0.3)	9	9	8	3.3	0.5	-
U2 - Beta(1, 1)	8.8	9	8.8	2.9	0.4	-
U3 - Beta(0.3, 0.7)	11.3	10.8	6.9	0.8	0.1	-
U3 - Beta(0.5, 0.5)	9.3	9.1	8.5	2.6	0.4	-
U3 - Beta(0.7, 0.3)	9.4	9	8	3.1	0.4	-
U3 - Beta(1, 1)	9.4	9.3	8.2	2.6	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	4.3	2.6	0.4	0	-
U2 - Beta(0.5, 0.5)	2.2	4	2.3	0.9	0.1	-
U2 - Beta(0.7, 0.3)	2.4	4	2	1	0.1	-
U2 - Beta(1, 1)	2.2	4	2.3	0.9	0.1	-
U3 - Beta(0.3, 0.7)	3.4	4.6	2.1	0.3	0	-
U3 - Beta(0.5, 0.5)	2.4	4.1	2.3	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.5	4	2.1	1	0.1	-
U3 - Beta(1, 1)	2.5	4.1	2.2	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1085.7
U2 - Beta(0.5, 0.5)						1119.8
U2 - Beta(0.7, 0.3)						1124.1
U2 - Beta(1, 1)						1121.2
U3 - Beta(0.3, 0.7)						1073.6
U3 - Beta(0.5, 0.5)						1115.1
U3 - Beta(0.7, 0.3)						1118.6
U3 - Beta(1, 1)						1113.5

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.1	28	42	4.8	0.3	10.8
U2 - Beta(0.5, 0.5)	19.2	28.2	33.6	7.6	0.1	11.3
U2 - Beta(0.7, 0.3)	18.4	31	31	8.7	0.1	10.8
U2 - Beta(1, 1)	17.3	27.9	34.7	8.8	0.1	11.2
U3 - Beta(0.3, 0.7)	18.5	29.5	36.7	3.9	0.2	11.2
U3 - Beta(0.5, 0.5)	23	30.6	28.3	7.1	0.1	10.9
U3 - Beta(0.7, 0.3)	24.5	29.6	26.5	8.1	0.1	11.2
U3 - Beta(1, 1)	23.7	29	27.8	8.3	0.2	11
Mean administrations						
U2 - Beta(0.3, 0.7)	8.1	9.5	9.1	1.8	0.1	-
U2 - Beta(0.5, 0.5)	8	8.7	8.3	3.2	0.4	-
U2 - Beta(0.7, 0.3)	8.1	8.8	7.8	3.4	0.6	-
U2 - Beta(1, 1)	7.8	8.8	8.4	3.2	0.4	-
U3 - Beta(0.3, 0.7)	8.9	9.7	8.3	1.4	0.1	-
U3 - Beta(0.5, 0.5)	8.4	8.8	8.1	2.9	0.4	-
U3 - Beta(0.7, 0.3)	8.4	8.9	7.6	3.3	0.4	-
U3 - Beta(1, 1)	8.4	8.9	7.9	3	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	4.3	2.9	0.7	0	-
U2 - Beta(0.5, 0.5)	1.9	4.1	2.3	1.1	0.1	-
U2 - Beta(0.7, 0.3)	2	4.2	2.1	1.1	0.1	-
U2 - Beta(1, 1)	1.8	4.2	2.3	1.1	0.1	-
U3 - Beta(0.3, 0.7)	2.4	4.4	2.7	0.6	0	-
U3 - Beta(0.5, 0.5)	2.1	4.2	2.2	1	0.1	-
U3 - Beta(0.7, 0.3)	2.1	4.3	2	1.1	0.1	-
U3 - Beta(1, 1)	2.1	4.2	2.2	1.1	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1024
U2 - Beta(0.5, 0.5)						1045.7
U2 - Beta(0.7, 0.3)						1050.7
U2 - Beta(1, 1)						1047.1
U3 - Beta(0.3, 0.7)						1015.7
U3 - Beta(0.5, 0.5)						1042.5
U3 - Beta(0.7, 0.3)						1044.1
U3 - Beta(1, 1)						1042

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.5	23.3	55.5	8.5	0.6	4.6
U2 - Beta(0.5, 0.5)	12.7	22.9	47.8	11.4	0.6	4.6
U2 - Beta(0.7, 0.3)	13.7	22.5	46.5	11.6	1	4.7
U2 - Beta(1, 1)	12.8	24.2	46.1	11.5	0.8	4.6
U3 - Beta(0.3, 0.7)	13.3	27.4	48.3	6.2	0.3	4.5
U3 - Beta(0.5, 0.5)	16.7	25.4	42	10.4	0.7	4.8
U3 - Beta(0.7, 0.3)	19.9	23.5	40.7	10.1	0.9	4.9
U3 - Beta(1, 1)	18.1	25.2	40.5	10.9	0.6	4.7
Mean administrations						
U2 - Beta(0.3, 0.7)	6.6	9.2	11.2	2.2	0.2	-
U2 - Beta(0.5, 0.5)	6.2	8	10.3	4	0.9	-
U2 - Beta(0.7, 0.3)	6.4	7.9	9.5	4.5	1.2	-
U2 - Beta(1, 1)	6.2	8	10.1	4.2	0.9	-
U3 - Beta(0.3, 0.7)	7.5	9.7	10.3	1.7	0.2	-
U3 - Beta(0.5, 0.5)	6.5	8.2	10.1	3.9	0.8	-
U3 - Beta(0.7, 0.3)	6.7	8.2	9.4	4.3	0.8	-
U3 - Beta(1, 1)	6.6	8.4	9.9	3.8	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.5	4.2	3.6	0.7	0.1	-
U2 - Beta(0.5, 0.5)	1.3	3.9	2.9	1.3	0.3	-
U2 - Beta(0.7, 0.3)	1.4	3.9	2.6	1.4	0.3	-
U2 - Beta(1, 1)	1.2	3.9	2.9	1.4	0.3	-
U3 - Beta(0.3, 0.7)	1.9	4.4	3.3	0.6	0	-
U3 - Beta(0.5, 0.5)	1.4	4	2.9	1.2	0.2	-
U3 - Beta(0.7, 0.3)	1.4	4	2.6	1.4	0.2	-
U3 - Beta(1, 1)	1.4	4.1	2.8	1.3	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1061.9
U2 - Beta(0.5, 0.5)						1089.9
U2 - Beta(0.7, 0.3)						1091.6
U2 - Beta(1, 1)						1089.1
U3 - Beta(0.3, 0.7)						1054.1
U3 - Beta(0.5, 0.5)						1084.5
U3 - Beta(0.7, 0.3)						1087.3
U3 - Beta(1, 1)						1084.6
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.9	47.2	7.4	0.4	0	36.1
U2 - Beta(0.5, 0.5)	9.4	44.5	8.6	0.5	0	37
U2 - Beta(0.7, 0.3)	9.6	43.4	9.1	0.7	0	37.2
U2 - Beta(1, 1)	9.4	43.6	9.2	0.5	0	37.3
U3 - Beta(0.3, 0.7)	8.6	49.2	5.6	0.3	0	36.3
U3 - Beta(0.5, 0.5)	9.5	44.8	8.6	0.2	0	36.9
U3 - Beta(0.7, 0.3)	9.6	44.8	7.9	0.6	0	37.1
U3 - Beta(1, 1)	10	44.6	7.9	0.5	0	37
Mean administrations						
U2 - Beta(0.3, 0.7)	8.5	13	3	0.5	0	-
U2 - Beta(0.5, 0.5)	8.5	11.1	4.1	1.2	0.2	-
U2 - Beta(0.7, 0.3)	8.5	10.9	4.1	1.5	0.2	-
U2 - Beta(1, 1)	8.5	11.2	4.1	1.2	0.2	-
U3 - Beta(0.3, 0.7)	8.7	13.3	2.7	0.4	0	-
U3 - Beta(0.5, 0.5)	8.7	11.3	3.9	1.1	0.1	-
U3 - Beta(0.7, 0.3)	8.7	10.9	4	1.3	0.2	-
U3 - Beta(1, 1)	8.7	11.2	4	1.1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	5.3	1	0.2	0	-
U2 - Beta(0.5, 0.5)	1.9	4.7	1.2	0.4	0	-
U2 - Beta(0.7, 0.3)	1.9	4.7	1.1	0.5	0.1	-
U2 - Beta(1, 1)	1.9	4.8	1.2	0.4	0.1	-
U3 - Beta(0.3, 0.7)	2	5.5	0.9	0.1	0	-
U3 - Beta(0.5, 0.5)	2	4.8	1.1	0.4	0	-
U3 - Beta(0.7, 0.3)	1.9	4.7	1.1	0.4	0.1	-
U3 - Beta(1, 1)	1.9	4.8	1.2	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						911.4
U2 - Beta(0.5, 0.5)						925.2
U2 - Beta(0.7, 0.3)						927.9
U2 - Beta(1, 1)						926.3
U3 - Beta(0.3, 0.7)						908.3
U3 - Beta(0.5, 0.5)						920.7
U3 - Beta(0.7, 0.3)						924.1
U3 - Beta(1, 1)						921.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	74.8	20	2.9	0	0	2.3
U2 - Beta(0.5, 0.5)	63.1	29.9	4.4	0.2	0	2.4
U2 - Beta(0.7, 0.3)	64.1	28.7	4.4	0.2	0	2.6
U2 - Beta(1, 1)	61.2	31.2	4.9	0.2	0	2.5
U3 - Beta(0.3, 0.7)	77.5	18.4	1.8	0	0	2.3
U3 - Beta(0.5, 0.5)	68.2	26.5	3	0.2	0	2.1
U3 - Beta(0.7, 0.3)	65.6	27.3	4.5	0.2	0	2.4
U3 - Beta(1, 1)	68.2	26.3	3	0.1	0	2.4
Mean administrations						
U2 - Beta(0.3, 0.7)	23.3	5.2	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	17.4	8.9	2.6	0.6	0	-
U2 - Beta(0.7, 0.3)	17	8.8	2.8	0.9	0.1	-
U2 - Beta(1, 1)	17.1	9	2.7	0.6	0	-
U3 - Beta(0.3, 0.7)	23.9	5	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	18.4	8.5	2.2	0.5	0	-
U3 - Beta(0.7, 0.3)	17.2	8.9	2.7	0.7	0.1	-
U3 - Beta(1, 1)	18.4	8.5	2.2	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.4	1.8	0.3	0	0	-
U2 - Beta(0.5, 0.5)	4.2	3.1	0.7	0.2	0	-
U2 - Beta(0.7, 0.3)	4.1	3.1	0.7	0.2	0	-
U2 - Beta(1, 1)	4.2	3.2	0.7	0.2	0	-
U3 - Beta(0.3, 0.7)	6.6	1.8	0.2	0	0	-
U3 - Beta(0.5, 0.5)	4.6	3	0.6	0.1	0	-
U3 - Beta(0.7, 0.3)	4.2	3.1	0.7	0.2	0	-
U3 - Beta(1, 1)	4.6	3	0.6	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1155.4
U2 - Beta(0.5, 0.5)						1177.7
U2 - Beta(0.7, 0.3)						1181.8
U2 - Beta(1, 1)						1177.6
U3 - Beta(0.3, 0.7)						1153.9
U3 - Beta(0.5, 0.5)						1173.2
U3 - Beta(0.7, 0.3)						1180.7
U3 - Beta(1, 1)						1171.4

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	9.4	16.6	56.3	9.5	0.3
U2 - Beta(0.5, 0.5)	17.4	12.1	12.4	43.7	14	0.4
U2 - Beta(0.7, 0.3)	20.9	13.7	12.4	39.8	12.8	0.4
U2 - Beta(1, 1)	19	12	11.7	42.3	14.6	0.4
U3 - Beta(0.3, 0.7)	15.1	13.5	17.1	49	4.9	0.4
U3 - Beta(0.5, 0.5)	25.2	14.7	10.6	39.2	9.8	0.5
U3 - Beta(0.7, 0.3)	27.6	15.6	11.7	34.1	10.6	0.4
U3 - Beta(1, 1)	28.4	14.8	11.9	35.5	9	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	5.5	5.9	7.2	9.7	1.7	-
U2 - Beta(0.5, 0.5)	4.3	4.8	5.9	10.3	4.7	-
U2 - Beta(0.7, 0.3)	4.5	4.9	6	9.5	5.1	-
U2 - Beta(1, 1)	4.3	4.9	5.9	10.3	4.6	-
U3 - Beta(0.3, 0.7)	6.2	6.5	7.1	8.9	1.2	-
U3 - Beta(0.5, 0.5)	4.5	4.9	6	10.3	4.2	-
U3 - Beta(0.7, 0.3)	4.7	5.1	6	9.5	4.6	-
U3 - Beta(1, 1)	4.7	5.1	5.8	10.2	4.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.2	3.1	2.7	3.2	0.6	-
U2 - Beta(0.5, 0.5)	0.5	2.9	1.9	3.4	1.3	-
U2 - Beta(0.7, 0.3)	0.6	3	1.9	3.2	1.4	-
U2 - Beta(1, 1)	0.5	3	1.9	3.3	1.3	-
U3 - Beta(0.3, 0.7)	1.5	3.4	2.7	3	0.4	-
U3 - Beta(0.5, 0.5)	0.6	3	1.9	3.4	1.2	-
U3 - Beta(0.7, 0.3)	0.7	3.1	1.9	3.1	1.3	-
U3 - Beta(1, 1)	0.7	3.1	1.8	3.3	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1052.8
U2 - Beta(0.5, 0.5)						1095.4
U2 - Beta(0.7, 0.3)						1096.9
U2 - Beta(1, 1)						1095.2
U3 - Beta(0.3, 0.7)						1042.4
U3 - Beta(0.5, 0.5)						1091.7
U3 - Beta(0.7, 0.3)						1090.3
U3 - Beta(1, 1)						1088.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.3	71.5	19	5.1	0.1	0
U2 - Beta(0.5, 0.5)	5.2	33.9	27.2	26.3	7.4	0
U2 - Beta(0.7, 0.3)	8	33.8	25.6	24.2	8.4	0
U2 - Beta(1, 1)	5.1	34.5	27.9	25.2	7.3	0
U3 - Beta(0.3, 0.7)	6.7	74.1	15.8	3.3	0.1	0
U3 - Beta(0.5, 0.5)	6.8	40.1	26.5	21.3	5.3	0
U3 - Beta(0.7, 0.3)	8.8	39.3	26.9	19.9	5.1	0
U3 - Beta(1, 1)	7.9	41.9	25.9	20.2	4.1	0
Mean administrations						
U2 - Beta(0.3, 0.7)	7.5	17.8	3.9	0.7	0.1	-
U2 - Beta(0.5, 0.5)	3.8	10.9	7.4	5.9	2.1	-
U2 - Beta(0.7, 0.3)	3.6	6.4	7.3	8.5	4.2	-
U2 - Beta(1, 1)	3.8	10.8	7.4	5.8	2.1	-
U3 - Beta(0.3, 0.7)	7.8	18.3	3.3	0.6	0	-
U3 - Beta(0.5, 0.5)	3.9	11.8	7.3	5.4	1.6	-
U3 - Beta(0.7, 0.3)	3.9	8	7.6	7.3	3.2	-
U3 - Beta(1, 1)	3.9	11.9	7.2	5.4	1.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	6.4	1.2	0.2	0	-
U2 - Beta(0.5, 0.5)	0.3	4.4	2.2	1.7	0.6	-
U2 - Beta(0.7, 0.3)	0.2	3	2.1	2.3	1.1	-
U2 - Beta(1, 1)	0.3	4.3	2.2	1.7	0.6	-
U3 - Beta(0.3, 0.7)	2.1	6.6	1.1	0.2	0	-
U3 - Beta(0.5, 0.5)	0.3	4.7	2.2	1.6	0.5	-
U3 - Beta(0.7, 0.3)	0.3	3.5	2.2	2	0.9	-
U3 - Beta(1, 1)	0.3	4.7	2.1	1.6	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1109.1
U2 - Beta(0.5, 0.5)						1153.1
U2 - Beta(0.7, 0.3)						1177.8
U2 - Beta(1, 1)						1154.2
U3 - Beta(0.3, 0.7)						1106.8
U3 - Beta(0.5, 0.5)						1147.5
U3 - Beta(0.7, 0.3)						1164.6
U3 - Beta(1, 1)						1147.2
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.3	8.2	41.7	34.7	12	0.1
U2 - Beta(0.5, 0.5)	14.4	13.6	34.1	25.7	12.1	0.1
U2 - Beta(0.7, 0.3)	15.4	14.8	33.4	25.4	10.9	0.1
U2 - Beta(1, 1)	16.8	14.8	30.9	25.6	11.8	0.1
U3 - Beta(0.3, 0.7)	7.6	11.3	42.5	30.5	8	0.1
U3 - Beta(0.5, 0.5)	19.3	17.7	31.8	23.4	7.7	0.1
U3 - Beta(0.7, 0.3)	19.6	17.3	32.3	22.7	8	0.1
U3 - Beta(1, 1)	23.6	17.4	30.4	21.3	7.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.2	5.6	11.1	6.8	2.3	-
U2 - Beta(0.5, 0.5)	3.5	4.2	8.5	8.5	5.3	-
U2 - Beta(0.7, 0.3)	3.6	4.4	8.3	8.3	5.4	-
U2 - Beta(1, 1)	3.6	4.4	8.5	8.4	5.2	-
U3 - Beta(0.3, 0.7)	4.4	6	11.3	6.6	1.8	-
U3 - Beta(0.5, 0.5)	3.6	4.5	8.7	8.5	4.7	-
U3 - Beta(0.7, 0.3)	3.8	4.7	8.5	8.2	4.9	-
U3 - Beta(1, 1)	3.8	4.8	8.6	8.4	4.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	3.2	4.1	2.5	0.8	-
U2 - Beta(0.5, 0.5)	0.2	2.8	2.8	3.2	1.7	-
U2 - Beta(0.7, 0.3)	0.3	2.9	2.7	3.2	1.7	-
U2 - Beta(1, 1)	0.2	2.9	2.8	3.2	1.7	-
U3 - Beta(0.3, 0.7)	0.7	3.3	4.2	2.4	0.6	-
U3 - Beta(0.5, 0.5)	0.3	2.9	2.9	3.2	1.5	-
U3 - Beta(0.7, 0.3)	0.3	3	2.8	3.1	1.5	-
U3 - Beta(1, 1)	0.4	3.1	2.8	3.2	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1033.7
U2 - Beta(0.5, 0.5)						1058.6
U2 - Beta(0.7, 0.3)						1060.6
U2 - Beta(1, 1)						1056.8
U3 - Beta(0.3, 0.7)						1029
U3 - Beta(0.5, 0.5)						1054.7
U3 - Beta(0.7, 0.3)						1056.5
U3 - Beta(1, 1)						1051.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.5	3.1	13.3	59.1	20.9	0.1
U2 - Beta(0.5, 0.5)	8.6	5.3	12.1	43.7	30.1	0.2
U2 - Beta(0.7, 0.3)	9.7	7.6	12.4	40.4	29.8	0.1
U2 - Beta(1, 1)	9.6	5.9	11.5	43.1	29.8	0.1
U3 - Beta(0.3, 0.7)	6.4	4	16.2	59.2	14.1	0.1
U3 - Beta(0.5, 0.5)	12.2	8.2	11.6	42.5	25.4	0.1
U3 - Beta(0.7, 0.3)	14.8	9.1	12.3	39.6	24.1	0.1
U3 - Beta(1, 1)	16	8.6	10.9	40.2	24.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.7	4.2	7.6	11.4	3.1	-
U2 - Beta(0.5, 0.5)	3.3	3.6	5.9	10.3	6.8	-
U2 - Beta(0.7, 0.3)	3.3	3.7	5.7	9.7	7.6	-
U2 - Beta(1, 1)	3.4	3.7	5.8	10.3	6.7	-
U3 - Beta(0.3, 0.7)	3.8	4.3	8	11.5	2.4	-
U3 - Beta(0.5, 0.5)	3.4	3.7	6	10.5	6.3	-
U3 - Beta(0.7, 0.3)	3.6	3.9	5.9	9.9	6.7	-
U3 - Beta(1, 1)	3.7	4	5.8	10.4	6.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	2.8	3.9	0.9	-
U2 - Beta(0.5, 0.5)	0.2	2.6	1.8	3.7	1.8	-
U2 - Beta(0.7, 0.3)	0.2	2.7	1.7	3.6	1.9	-
U2 - Beta(1, 1)	0.2	2.7	1.8	3.7	1.8	-
U3 - Beta(0.3, 0.7)	0.4	2.8	3	3.9	0.7	-
U3 - Beta(0.5, 0.5)	0.2	2.7	1.8	3.8	1.7	-
U3 - Beta(0.7, 0.3)	0.3	2.8	1.8	3.6	1.7	-
U3 - Beta(1, 1)	0.3	2.8	1.7	3.7	1.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1061.5
U2 - Beta(0.5, 0.5)						1093.3
U2 - Beta(0.7, 0.3)						1098.3
U2 - Beta(1, 1)						1093.5
U3 - Beta(0.3, 0.7)						1055.3
U3 - Beta(0.5, 0.5)						1091.5
U3 - Beta(0.7, 0.3)						1089.6
U3 - Beta(1, 1)						1087
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.6	11	51.7	21.8	4.7	4.2
U2 - Beta(0.5, 0.5)	12.4	12.2	43.8	22.8	5.2	3.6
U2 - Beta(0.7, 0.3)	14.2	12.4	41.3	23.2	5.4	3.5
U2 - Beta(1, 1)	14	11	44.1	23	4.8	3.1
U3 - Beta(0.3, 0.7)	11.8	12.6	50.4	18.3	2.6	4.3
U3 - Beta(0.5, 0.5)	15.7	15.1	41.8	21.3	2.8	3.3
U3 - Beta(0.7, 0.3)	16.8	16.1	38.7	21.6	3.5	3.3
U3 - Beta(1, 1)	19.6	13.6	39	21.7	3.1	3
Mean administrations						
U2 - Beta(0.3, 0.7)	5	6.5	12.2	4.9	1.1	-
U2 - Beta(0.5, 0.5)	4.9	6.1	9.6	6.3	2.8	-
U2 - Beta(0.7, 0.3)	4.9	6.1	9.4	6.4	3	-
U2 - Beta(1, 1)	5	6.2	9.6	6.3	2.7	-
U3 - Beta(0.3, 0.7)	5.4	6.7	12.3	4.5	0.8	-
U3 - Beta(0.5, 0.5)	5.2	6.4	9.6	6.2	2.4	-
U3 - Beta(0.7, 0.3)	5.4	6.4	9.1	6.3	2.6	-
U3 - Beta(1, 1)	5.7	6.5	9.2	6	2.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.5	4.1	1.7	0.4	-
U2 - Beta(0.5, 0.5)	0.8	3.5	2.9	2.2	0.9	-
U2 - Beta(0.7, 0.3)	0.8	3.5	2.9	2.2	0.9	-
U2 - Beta(1, 1)	0.9	3.5	2.9	2.2	0.9	-
U3 - Beta(0.3, 0.7)	1.1	3.5	4.1	1.6	0.3	-
U3 - Beta(0.5, 0.5)	1	3.6	2.9	2.1	0.8	-
U3 - Beta(0.7, 0.3)	1.1	3.6	2.8	2.2	0.8	-
U3 - Beta(1, 1)	1.2	3.7	2.8	2.1	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1052.1
U2 - Beta(0.5, 0.5)						1066
U2 - Beta(0.7, 0.3)						1067.1
U2 - Beta(1, 1)						1064.6
U3 - Beta(0.3, 0.7)						1047
U3 - Beta(0.5, 0.5)						1062.6
U3 - Beta(0.7, 0.3)						1061.5
U3 - Beta(1, 1)						1056.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.3	64.1	24.4	4.6	0.5	0.1
U2 - Beta(0.5, 0.5)	9.1	39.7	32.5	15	3.6	0.1
U2 - Beta(0.7, 0.3)	11.6	38.1	30.7	16.4	3.1	0.1
U2 - Beta(1, 1)	11.1	38.6	32.3	14.5	3.4	0.1
U3 - Beta(0.3, 0.7)	9.2	67.5	20.4	2.6	0.2	0.1
U3 - Beta(0.5, 0.5)	14	45.1	27.6	10.7	2.5	0.1
U3 - Beta(0.7, 0.3)	14.8	43.1	25.8	13.8	2.4	0.1
U3 - Beta(1, 1)	15.1	45.6	26.8	10	2.4	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	16.6	4.8	0.8	0.1	-
U2 - Beta(0.5, 0.5)	4.4	10.9	8.5	4.6	1.6	-
U2 - Beta(0.7, 0.3)	4.4	8.7	8.7	5.6	2.6	-
U2 - Beta(1, 1)	4.4	10.9	8.5	4.6	1.5	-
U3 - Beta(0.3, 0.7)	8	17	4.3	0.6	0.1	-
U3 - Beta(0.5, 0.5)	4.6	12	8.1	4.1	1.2	-
U3 - Beta(0.7, 0.3)	4.6	10	8.3	5.2	1.8	-
U3 - Beta(1, 1)	4.6	12	8	4	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	6.2	1.5	0.2	0	-
U2 - Beta(0.5, 0.5)	0.5	4.4	2.4	1.3	0.5	-
U2 - Beta(0.7, 0.3)	0.5	3.8	2.2	1.6	0.7	-
U2 - Beta(1, 1)	0.5	4.4	2.3	1.3	0.4	-
U3 - Beta(0.3, 0.7)	2.2	6.3	1.3	0.2	0	-
U3 - Beta(0.5, 0.5)	0.6	4.8	2.2	1.2	0.3	-
U3 - Beta(0.7, 0.3)	0.6	4.2	2.1	1.5	0.5	-
U3 - Beta(1, 1)	0.6	4.8	2.2	1.2	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1103.3
U2 - Beta(0.5, 0.5)						1155.2
U2 - Beta(0.7, 0.3)						1169.8
U2 - Beta(1, 1)						1155.2
U3 - Beta(0.3, 0.7)						1099.6
U3 - Beta(0.5, 0.5)						1148.8
U3 - Beta(0.7, 0.3)						1160
U3 - Beta(1, 1)						1146.3
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9	40.6	17.4	9.4	1.9	21.7
U2 - Beta(0.5, 0.5)	12	41.2	16.1	8.2	1.6	20.9
U2 - Beta(0.7, 0.3)	13	41	16	8.7	1.6	19.7
U2 - Beta(1, 1)	14.7	39.7	14.7	8.4	1.4	21.1
U3 - Beta(0.3, 0.7)	11.5	42.8	15.3	6.6	1.5	22.3
U3 - Beta(0.5, 0.5)	16.1	41.3	13	7.1	1.1	21.4
U3 - Beta(0.7, 0.3)	17.3	41.1	12.8	6.5	1.1	21.2
U3 - Beta(1, 1)	18.7	39.9	12.2	6.6	0.9	21.7
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	11.2	5.4	2.6	0.6	-
U2 - Beta(0.5, 0.5)	7.5	9.8	5.6	3.5	1.2	-
U2 - Beta(0.7, 0.3)	7.4	9.6	5.5	3.8	1.4	-
U2 - Beta(1, 1)	7.4	9.9	5.6	3.5	1.2	-
U3 - Beta(0.3, 0.7)	8.1	11.7	5.1	2.2	0.4	-
U3 - Beta(0.5, 0.5)	7.7	10	5.4	3.4	1.1	-
U3 - Beta(0.7, 0.3)	7.7	10	5.4	3.5	1.1	-
U3 - Beta(1, 1)	7.9	10	5.3	3.4	1.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	5.2	1.8	1	0.2	-
U2 - Beta(0.5, 0.5)	1.6	4.8	1.7	1.3	0.4	-
U2 - Beta(0.7, 0.3)	1.6	4.7	1.6	1.4	0.4	-
U2 - Beta(1, 1)	1.6	4.8	1.7	1.3	0.4	-
U3 - Beta(0.3, 0.7)	2	5.3	1.8	0.8	0.1	-
U3 - Beta(0.5, 0.5)	1.7	4.9	1.7	1.2	0.4	-
U3 - Beta(0.7, 0.3)	1.7	4.9	1.7	1.3	0.3	-
U3 - Beta(1, 1)	1.8	4.9	1.6	1.2	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						962.7
U2 - Beta(0.5, 0.5)						978.2
U2 - Beta(0.7, 0.3)						979.5
U2 - Beta(1, 1)						980.3
U3 - Beta(0.3, 0.7)						960.3
U3 - Beta(0.5, 0.5)						973.3
U3 - Beta(0.7, 0.3)						976.7
U3 - Beta(1, 1)						972.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.4	48.3	16.1	4.6	0.8	18.8
U2 - Beta(0.5, 0.5)	13.1	47.9	16.3	3.9	0.8	18
U2 - Beta(0.7, 0.3)	13.1	47.6	17.2	4.7	0.7	16.7
U2 - Beta(1, 1)	15.1	45.6	15.8	5.2	0.7	17.6
U3 - Beta(0.3, 0.7)	15.8	49.5	13.1	2.8	0.5	18.3
U3 - Beta(0.5, 0.5)	17.1	48.7	14	2.4	0.6	17.2
U3 - Beta(0.7, 0.3)	17	47.8	14	3.3	0.6	17.3
U3 - Beta(1, 1)	17.9	48	13	2.9	0.4	17.8
Mean administrations						
U2 - Beta(0.3, 0.7)	7.6	12.4	5.5	2.3	0.5	-
U2 - Beta(0.5, 0.5)	7.5	11.1	5.7	3.1	0.9	-
U2 - Beta(0.7, 0.3)	7.4	10.9	5.8	3.2	1	-
U2 - Beta(1, 1)	7.5	11.1	5.7	3	0.9	-
U3 - Beta(0.3, 0.7)	7.9	12.8	5.2	2	0.4	-
U3 - Beta(0.5, 0.5)	7.8	11.2	5.5	2.9	0.8	-
U3 - Beta(0.7, 0.3)	7.8	11.1	5.6	3	0.8	-
U3 - Beta(1, 1)	7.9	11.3	5.4	2.9	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	5.6	1.9	0.9	0.2	-
U2 - Beta(0.5, 0.5)	1.6	5.3	1.8	1.1	0.3	-
U2 - Beta(0.7, 0.3)	1.6	5.3	1.8	1.2	0.3	-
U2 - Beta(1, 1)	1.6	5.3	1.8	1.1	0.3	-
U3 - Beta(0.3, 0.7)	1.9	5.7	1.8	0.8	0.1	-
U3 - Beta(0.5, 0.5)	1.8	5.4	1.7	1.1	0.3	-
U3 - Beta(0.7, 0.3)	1.8	5.3	1.7	1.1	0.3	-
U3 - Beta(1, 1)	1.8	5.4	1.7	1.1	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						983.3
U2 - Beta(0.5, 0.5)						993.4
U2 - Beta(0.7, 0.3)						995
U2 - Beta(1, 1)						992.7
U3 - Beta(0.3, 0.7)						981.4
U3 - Beta(0.5, 0.5)						992.2
U3 - Beta(0.7, 0.3)						990.8
U3 - Beta(1, 1)						989

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.